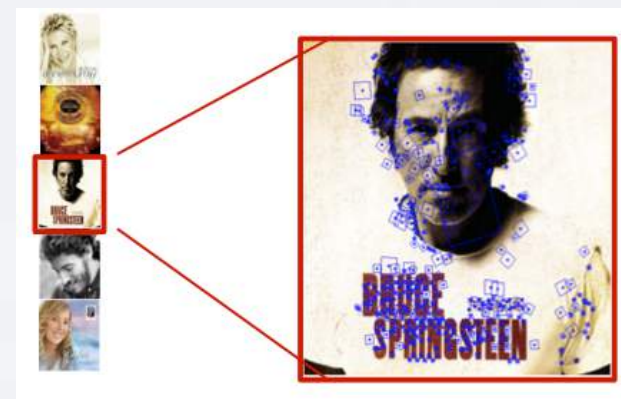
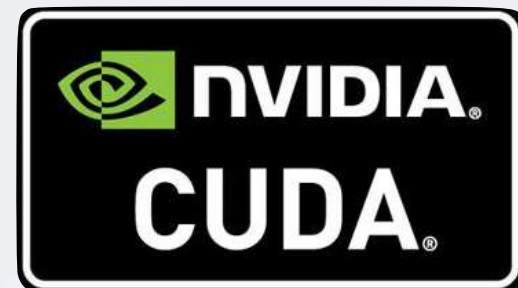
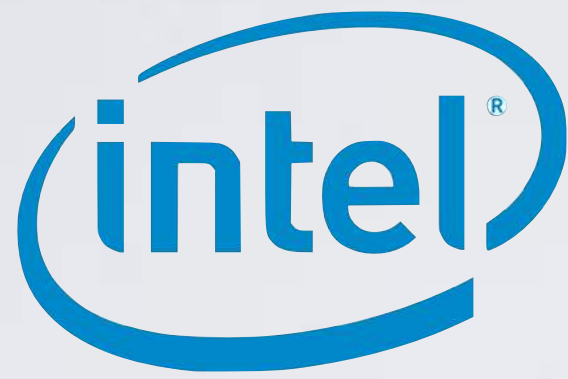


Eiffel Tower
Robert Delaunay



Abstraction or Instantiation on Building the Internet for Smarter Places

A Keynote for PCaPAC 2018
by Wei-Chao Chen @ Skywatch
weichao.chen@skywatch24.com



National Taiwan University

1998

2002

2006

2010

2014

2018

Wei-Chao Chen | 陳維超 | weichao.chen@skywatch24.com

PLATFORM

"the Internet"



Photo by DAVID ILIFF. License: CC-BY-SA 3.0

VEHICLE

“the Things”



Photo Credit: Jaguar Land Rover



INTERNET OF THINGS?

Photo Credit: Top Gear, BBC

HORIZONTAL THINKING ABSTRACT

Meta Apps

Create standards & platform

Application

and ask people to follow

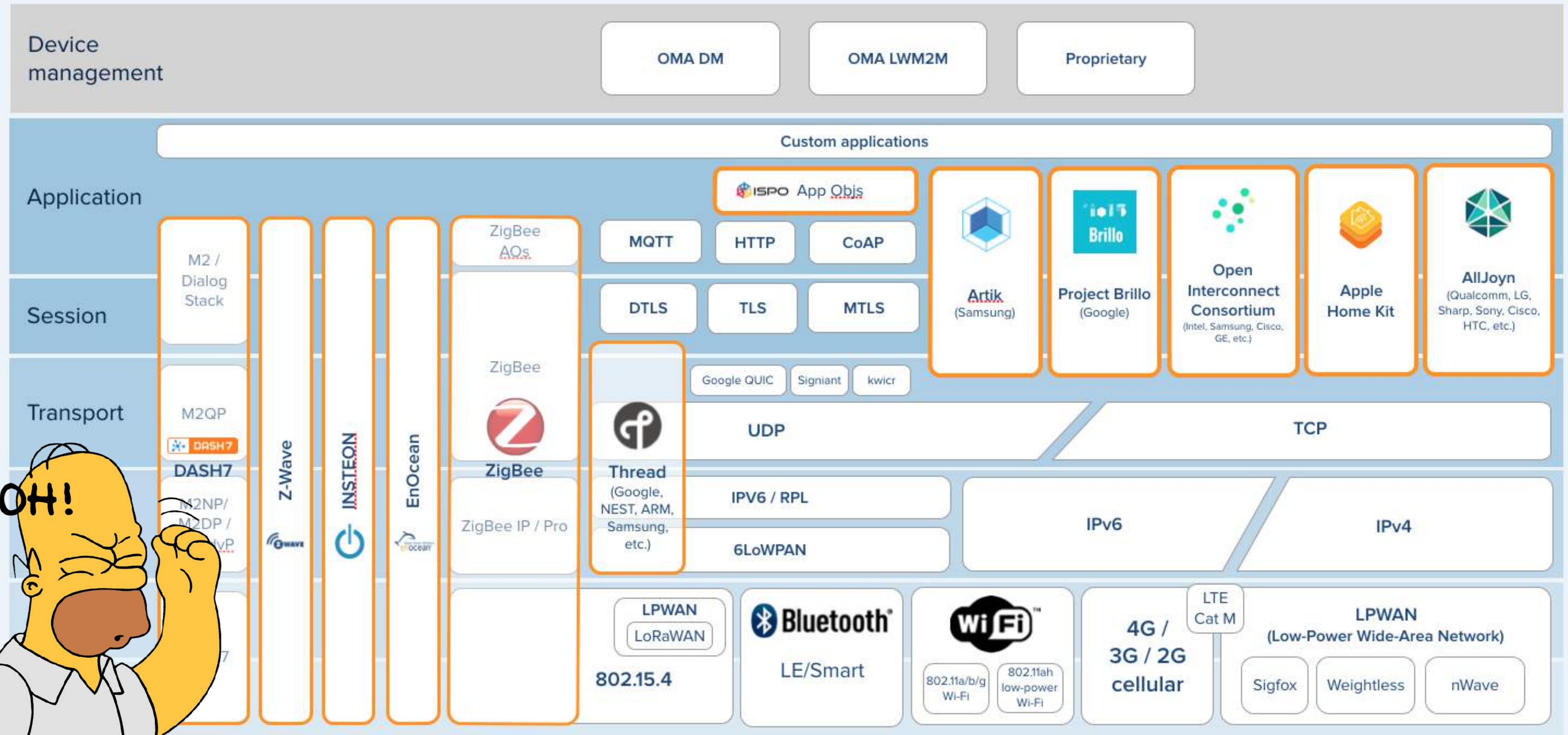
Transport

but this can become

Connectivity

too aggressive and limit spirits of innovation

Common protocols and standards used in IoT systems



Celebrating diversity, or fighting for monopoly?

Created by Dan Ledger, @dledge

v0.9

WHO WON?

Beta
B



VHS



VCD in China
DVD in a few years

Content providers?

Application Intelligence



Streaming



Integration

Microsoft Azure

ADVANTECH
WISE-PaaS



Analytics



Prediction

RESTful APIs

IoT Device Management

WISE-PaaS/RMM

IoT Data and Device
Management Platform



Data Storage



Device Management



Data Flow
Logic Designer



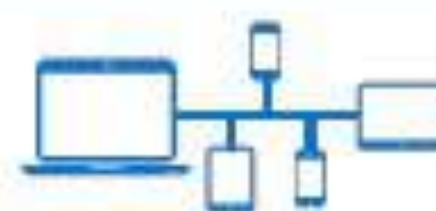
Dashboard Builder

MQTT

Data Acquisition & Transition

WISE-Agent

Gateway Connectivity SDK



Protocol Handler



MQTT Connector



Device Monitoring

Wireless Gateway

Multiple I/O Gateway

INDUSTRY 4.0

INTRANET-OF-THINGS?



Data privacy is important
Proprietary data interface
Legacy equipment
PLM, MES, M2M, ERP

Consumer technology
trickle back to factory?!

Siemens' state-of-the-art Electronic Works facility in Amberg, Germany, integrates its manufacturing, production and automation systems to process 1.6 billion components from 250 suppliers with 99% reliability.

<https://www.industryweek.com/technology/dawn-smart-factory>

VERTICAL THINKING INSTANCE

Meta Apps

Life is too short to wait for an uber-standard, so...

Application

Transport

Connectivity

Make useful

things that are,

unfortunately

vertical & isolated



PHILIPS









APP HELL...

iPhone Apps



L5 Remote
Utilities

+ Free ▾



C4Home
Utilities

Free ▾



Sweet Home
Automation for...
Lifestyle

Free ▾



SMART G4 Home
Automation BUS
Lifestyle

Free ▾



iSmart At Home -
Controller Version
Utilities

Free ▾



BeoLink
Utilities

+ Free ▾



Crestron Mobile
Lifestyle

Free ▾



Inspire Home
Automation
Utilities

+ Free ▾



InControl Home
Automation
Lifestyle

+ Free ▾



Appaydium Digital
Home Automation
Lifestyle

+ Free ▾



Control4 MyHome
- iPhone version
Lifestyle

Free ▾



RedEye
Entertainment

+ Free ▾



Home Automation
Lifestyle

Free ▾



L5 Remote SD
Utilities

Free ▾



RedEye mini
Entertainment

+ Free ▾



uControl Smart
Home Automation
Lifestyle

+ Free ▾



Switch
Automation.
Lifestyle

+ Free ▾



iViewer 4
Lifestyle

+ Free ▾

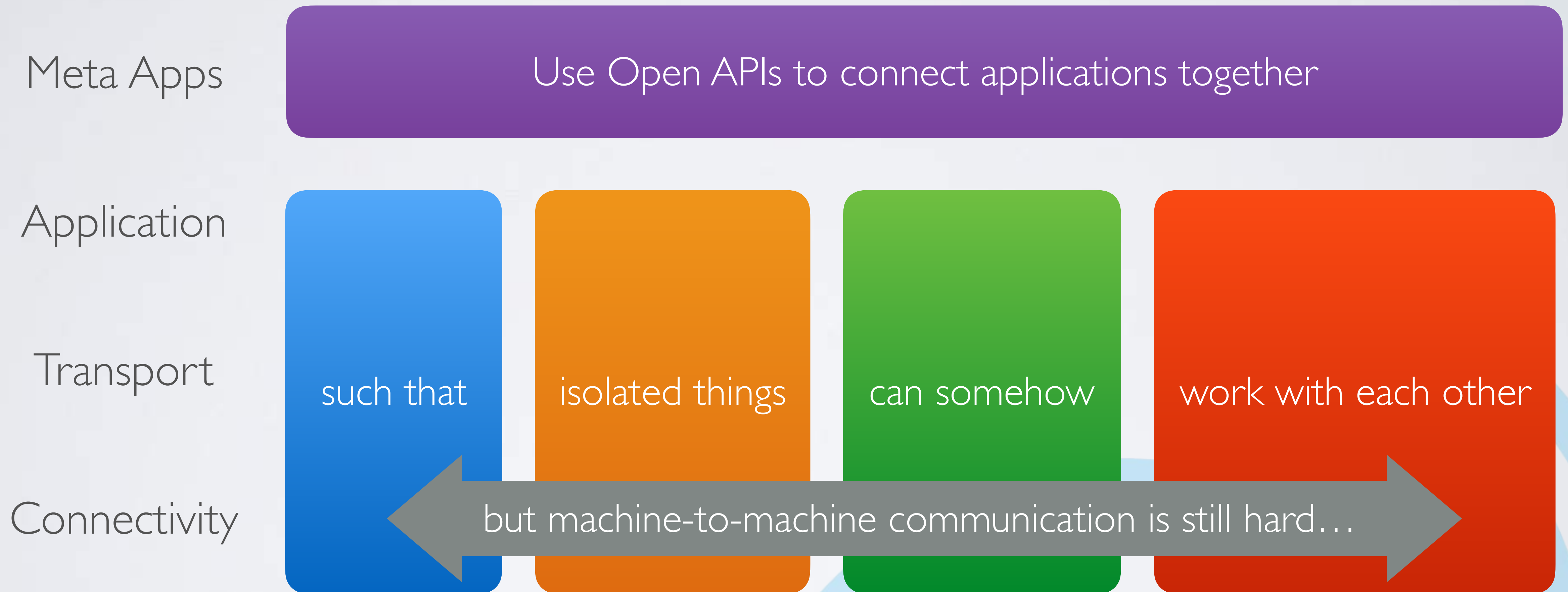
Source: iTunes



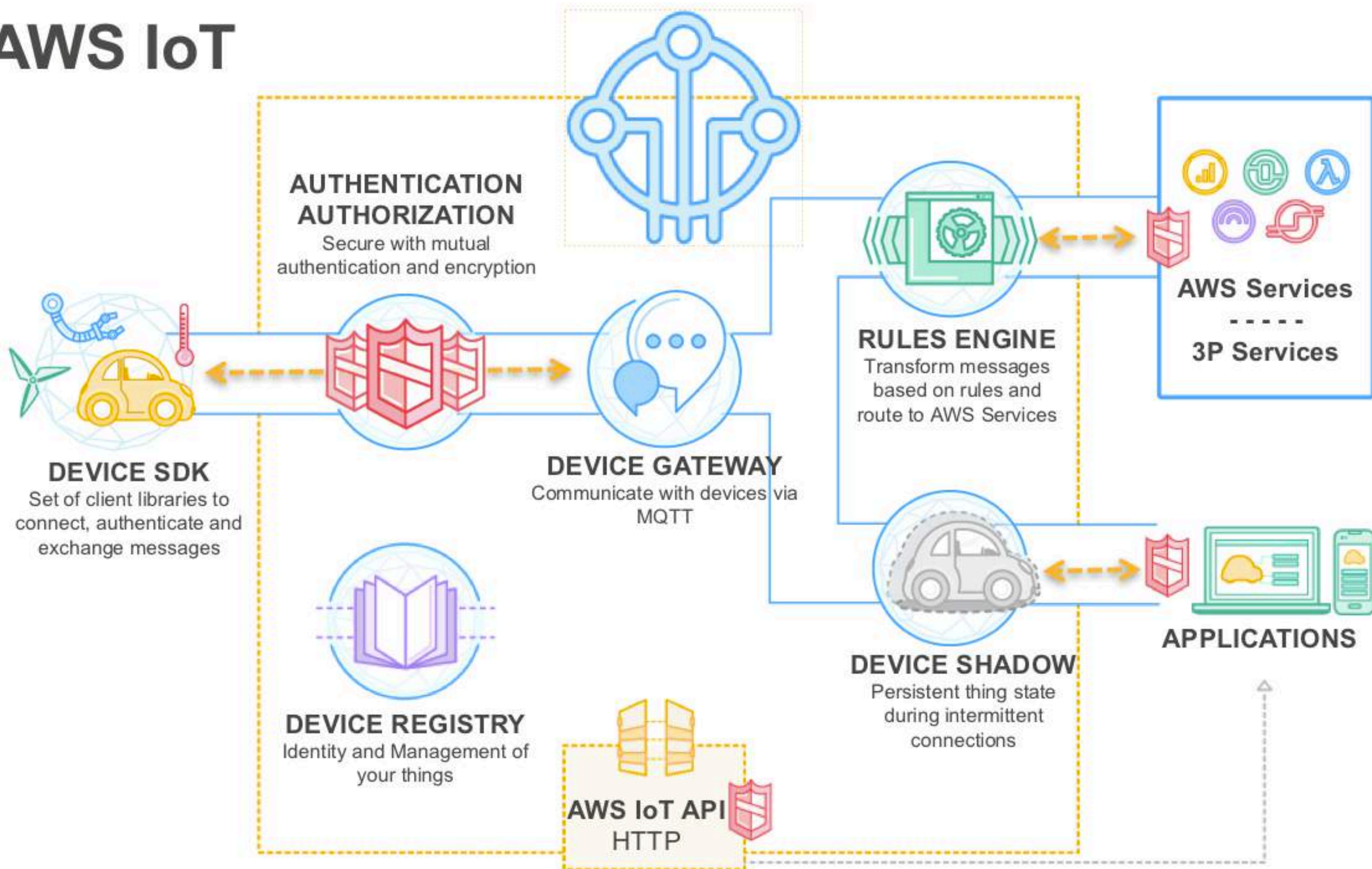
GOLDBERG: 1 – NEWTON: 0



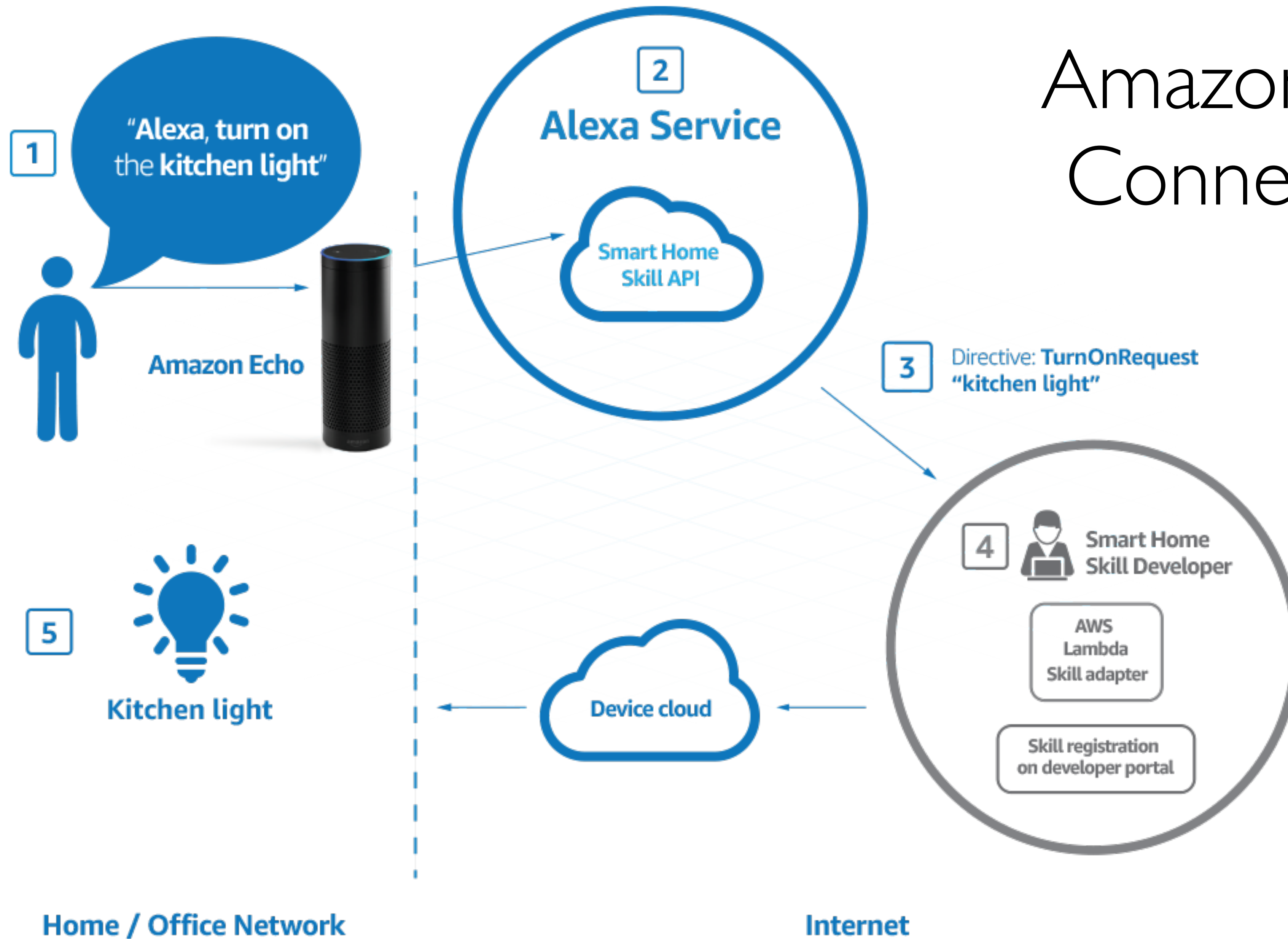
API ECONOMY



AWS IoT



Amazon Echo Connectivity



amazon echo

Always ready, connected, and fast. **Just ask.**



Too Harmless?

"Alexa, play 'Today's Hits' on Pandora."

"Alexa, play the station KEXP."

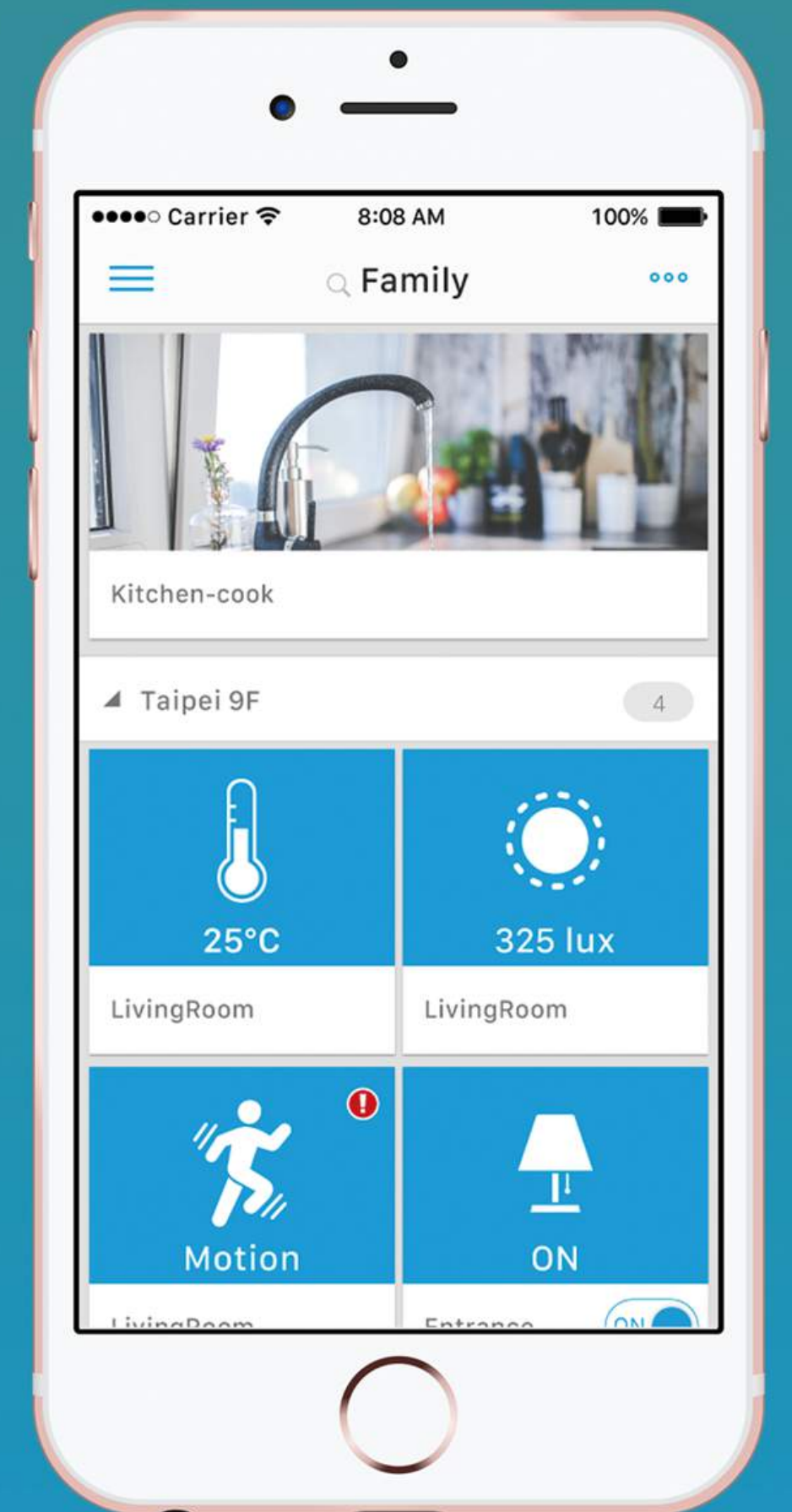
"Alexa, find me a Chinese restaurant."

"Alexa, set a timer for 20 minutes."

"Alexa, turn on the lights."

The Skywatch logo, featuring a stylized white eye icon with radiating lines and the word "Skywatch" in a white, rounded, sans-serif font, set against a blue background.

for smart places





single
site

达芙妮 DAPHNE
ALDO
SHOES & ACCESSORIES

200
stores



our initial customers
single shops to chain stores



2 shops



20
stores

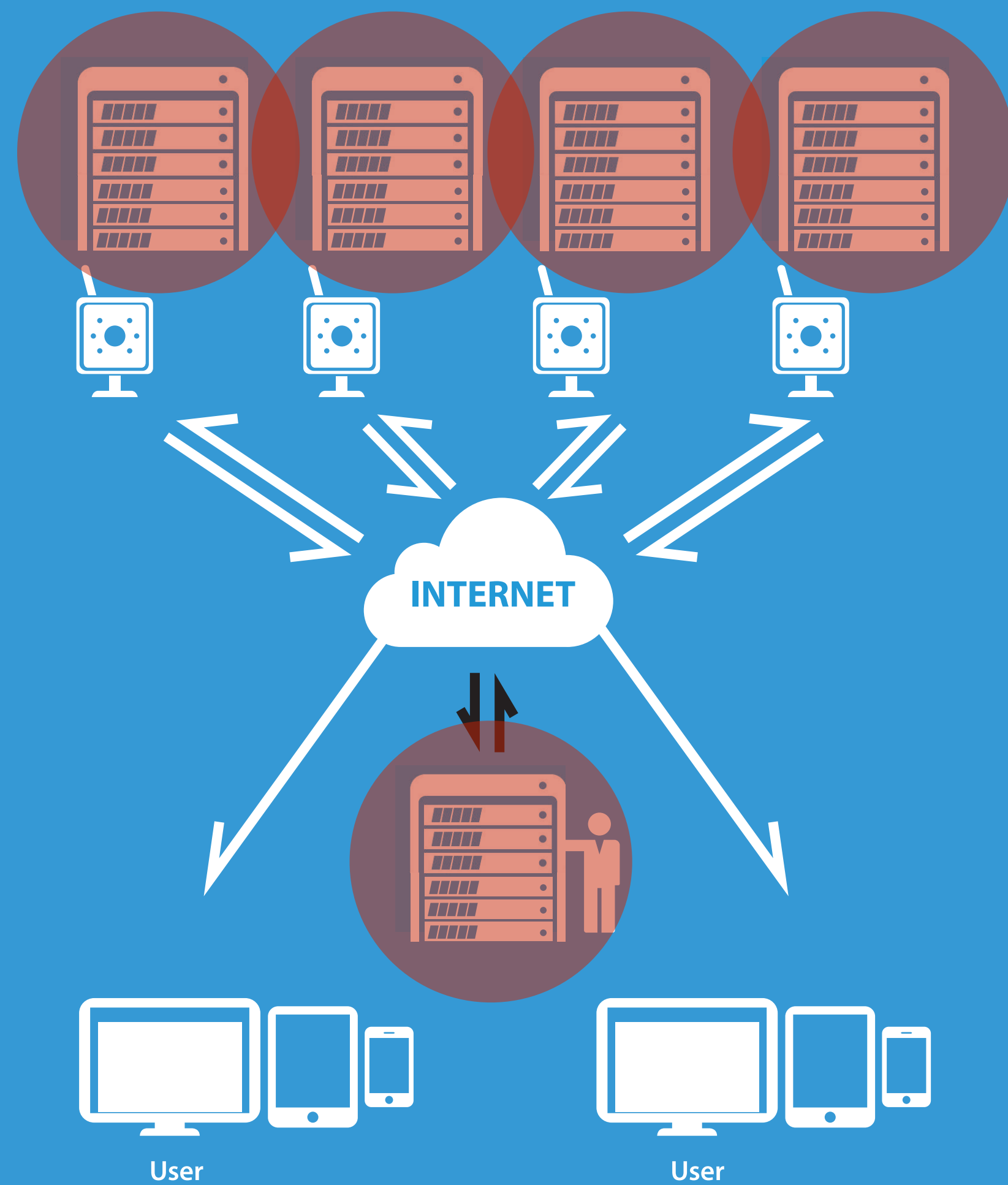
PANDORA
UNFORGETTABLE MOMENTS

Cash Flow

Regulations

Personnel

Security



Security Systems

expensive to build & manage

Security Centers

information overload



A Better Security System

**For
Users**

Less hardware
Easier to install
Lower cost

**For
Operators**

Better alarm accuracy
Lower personnel cost
Cheaper to operate

**For
the World**

Prevent crimes
Keep the jails empty
Makes the world safer

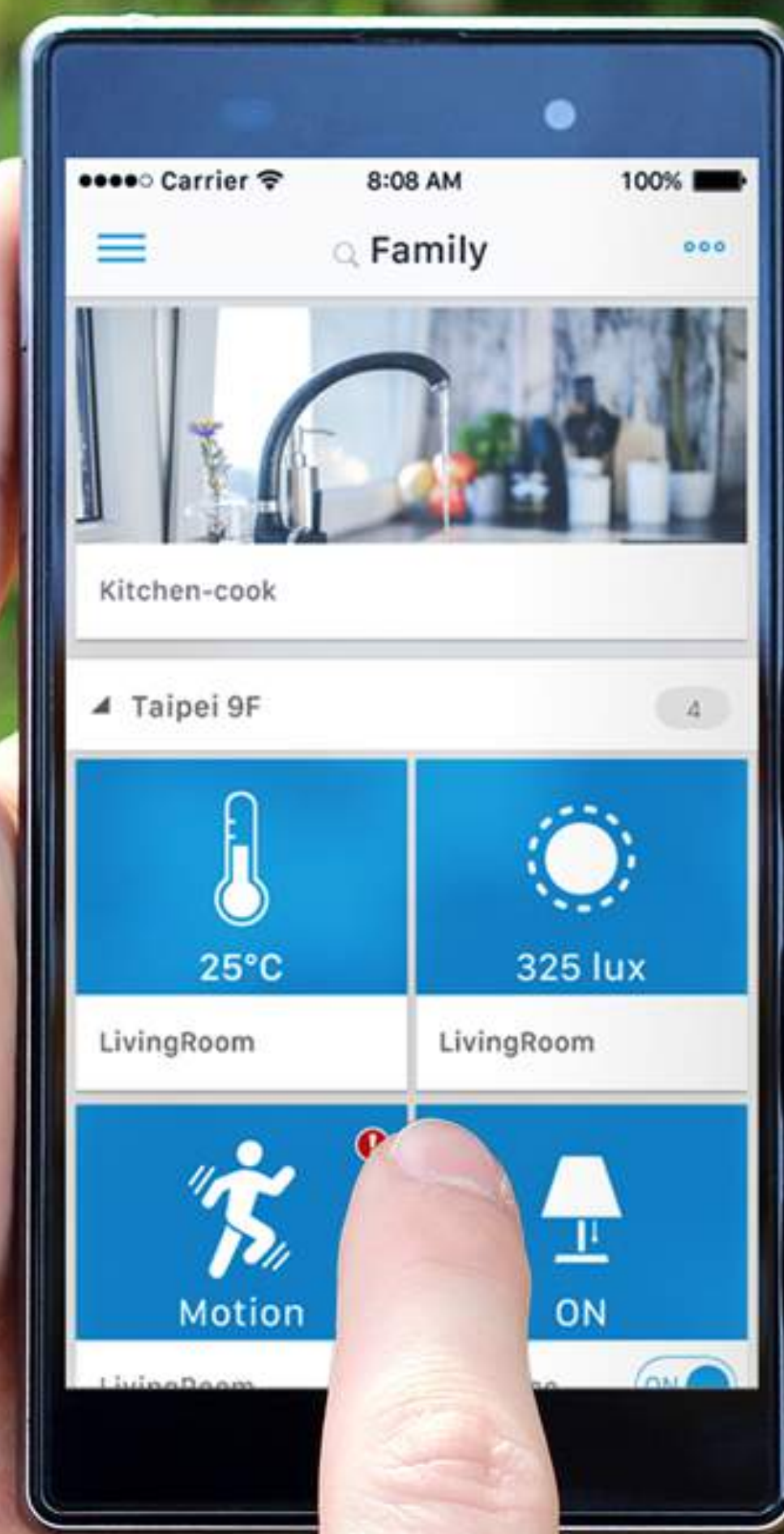
A Better Security System

For
Users

For
Operators

For
the World

a SaaS with connected devices and AI

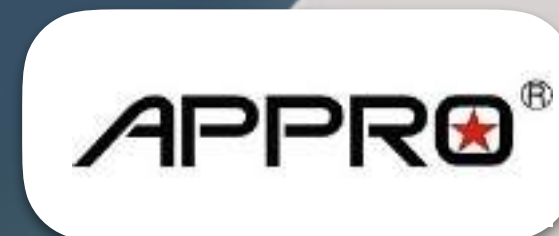


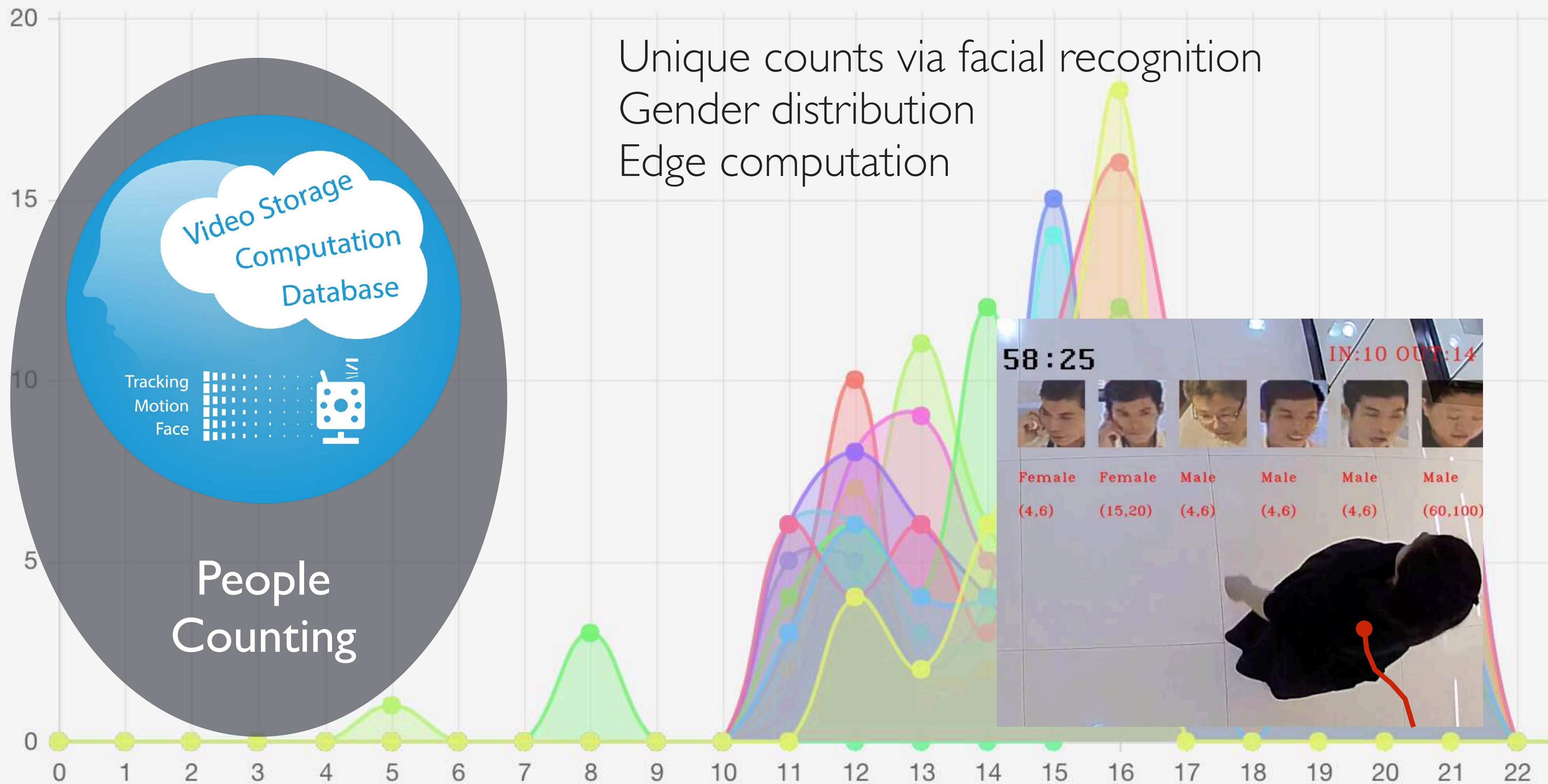
Simple setup
Multiple locations
Cloud recording
P2P/relay connectivity

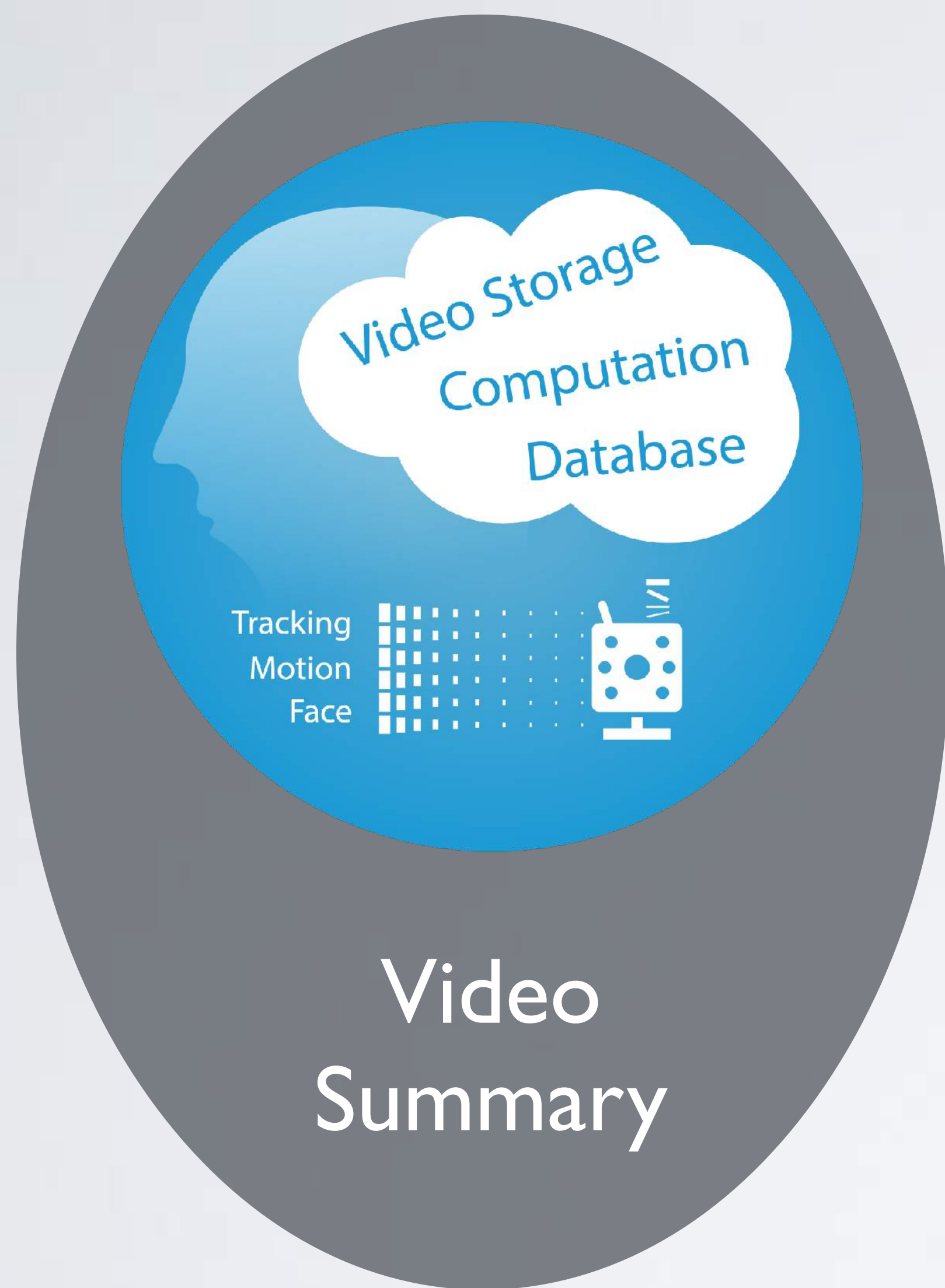
Compatibility



Supports major hardware vendors
One-stop integration
Open API economy







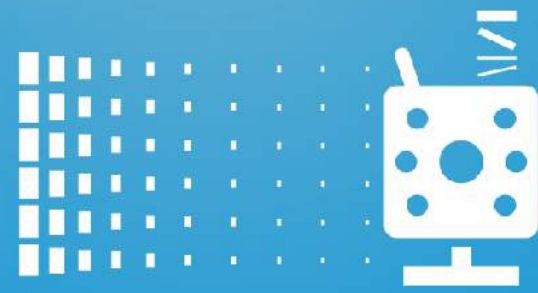
Compressed domain analysis
Extremely fast algorithm



12-28-2016 Wed 19:35:22

Video Storage
Computation
Database

Tracking
Motion
Face

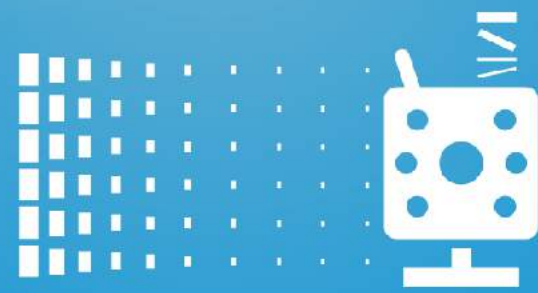


Video
Summary

Hallway

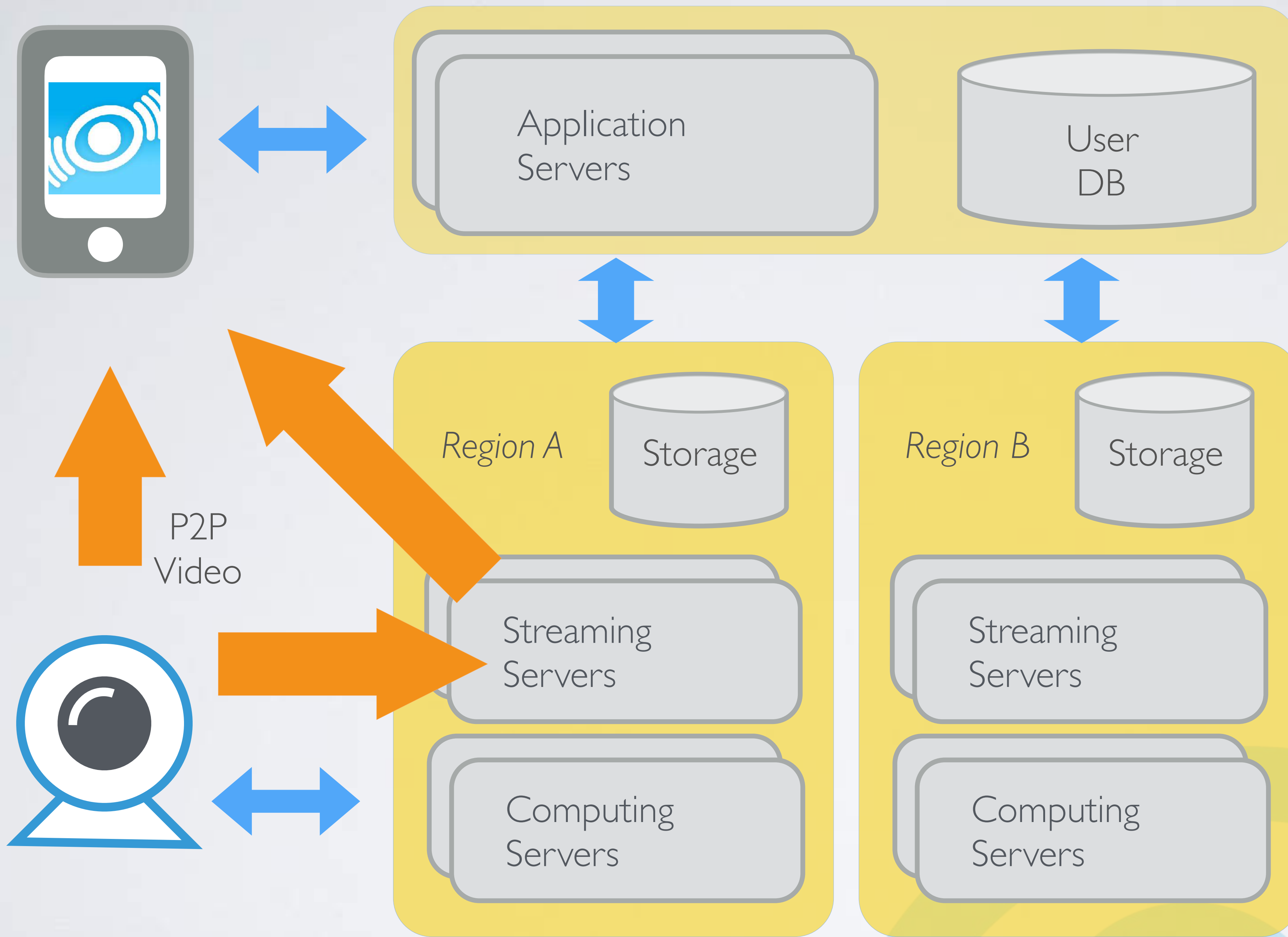
Video Storage
Computation
Database

Tracking
Motion
Face

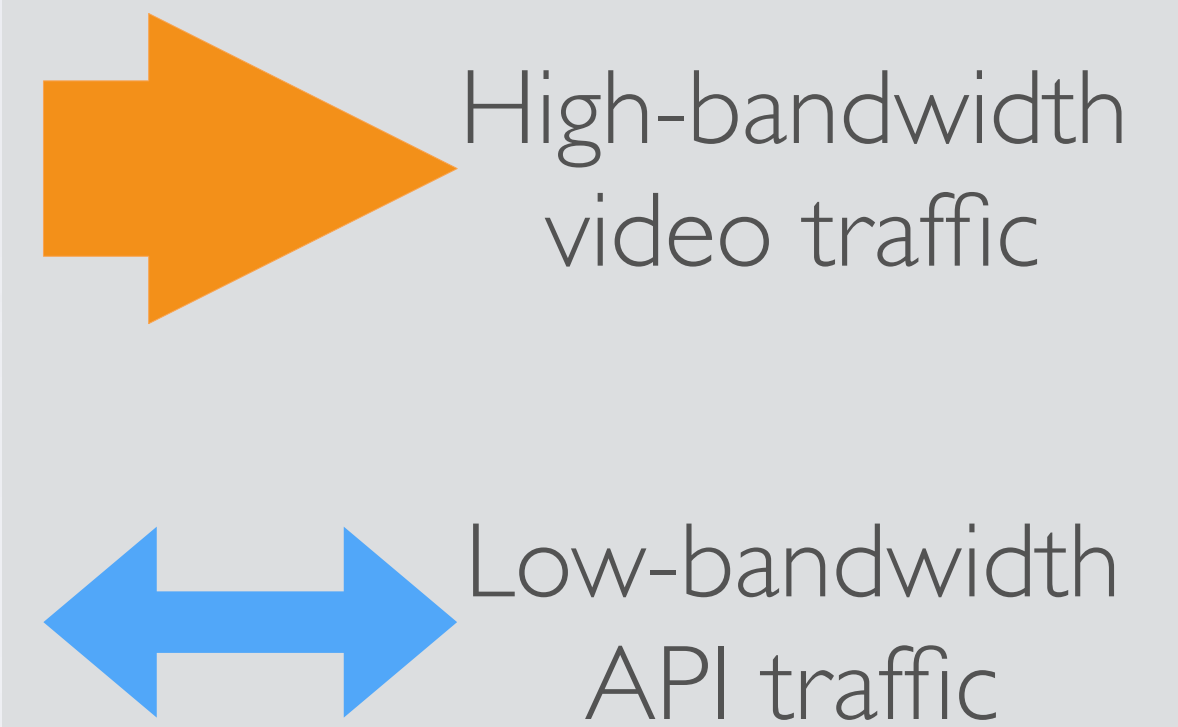


Automatic
Caption

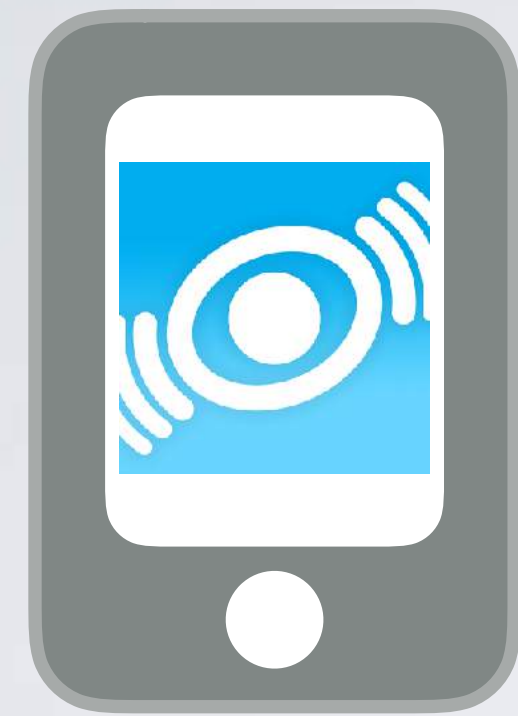
I saw...A man is doing stunts.



Streaming Architecture Overview



IoT Architecture Overview



Open API

Application
Servers

User DB

HTTP traffic



Z-Wave
Zigbee

MQTT
traffic

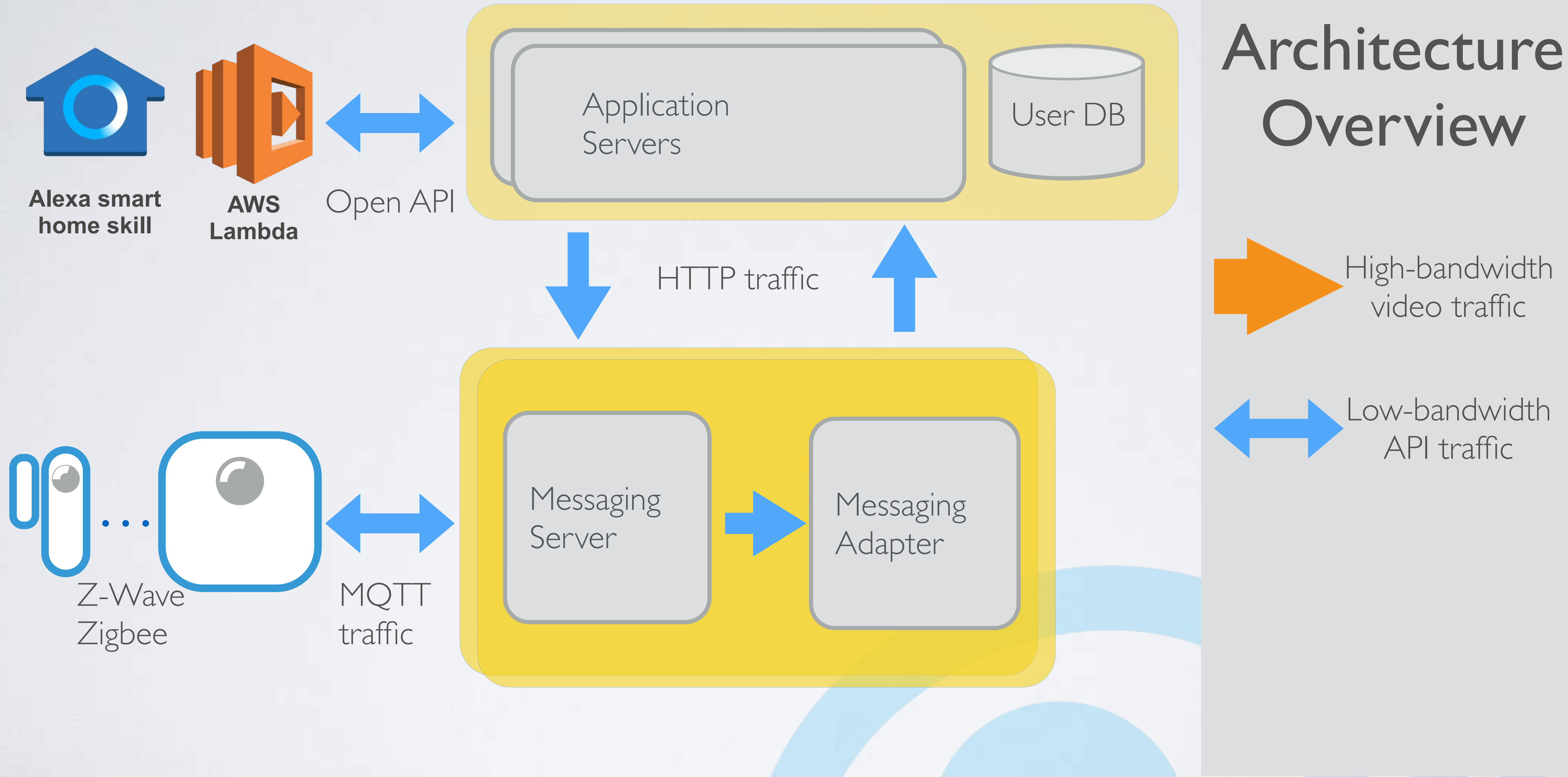
Messaging
Server

Messaging
Adapter

High-bandwidth
video traffic

Low-bandwidth
API traffic

IoT Architecture Overview



Channel Partners

Direct
Customers

Operators

Hardware &
Software SI

ZENSHIN
Systems Corporation

IPTec

TSOLUTIONS
Trustworthy Distributor

SOLUCIONES INTEGRALES
INFOKRAUSE
Ampliamos su visión.
Aseguramos su vida

OPTIMA
NETWORKS

SUMO

Taiwan Mobile
台灣大哥大

SKS
新光保全

中華電信
Chunghwa Telecom

FAREASTONE
遠傳

VIVOTEK
BUILT WITH RELIABILITY

Philic

TUTK

HIKVISION

Device cloud
opportunities

Use Case: Kbro Home Security Service Cloud

- Monthly rental model
- Telecom cloud center
- 3rd party hardware vendor
- Skywatch powered



HOME SECURITY
居 · 家 · 防 · 護

24小時全年無休守護你的家
優惠價\$100/月

全新推出

智慧縮時錄影功能
快速找到你要的錄影畫面

還在花時間找畫面? HomeSecurity 居家防護首創智慧縮時功能幫你自動省略靜止畫面，並保留所有人事物的動態畫面，讓你輕鬆調閱查看歷史影像，方便省時又安心!



一通電話到府安裝
24小時錄影安心有保障

影像自動儲存雲端
省時快轉30天不必等

首創智慧縮時錄影
立即快速調閱動態影像

遠端手機APP觀看
簡單操作監看多螢畫面

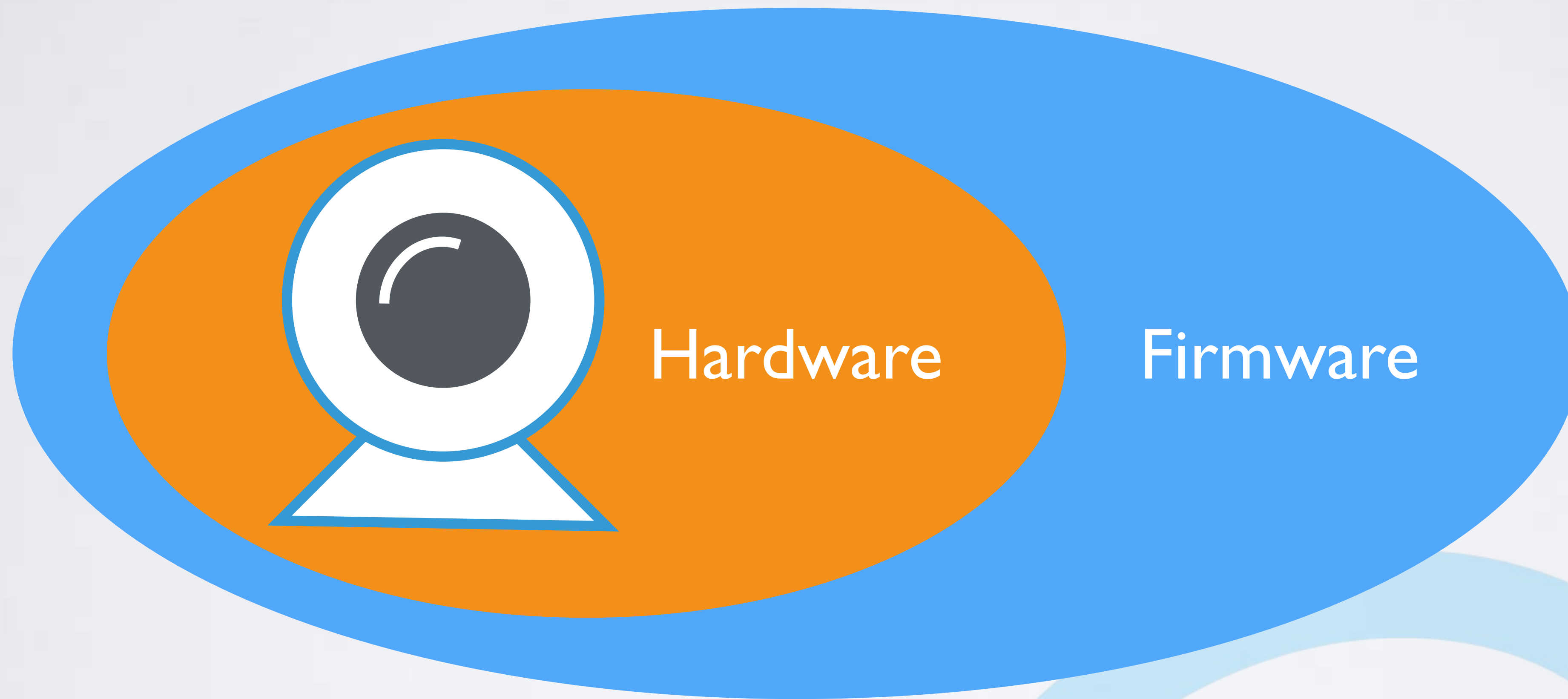
From Service Silo to Open Connectivity

Open APIs enable services to break out of vertical silos



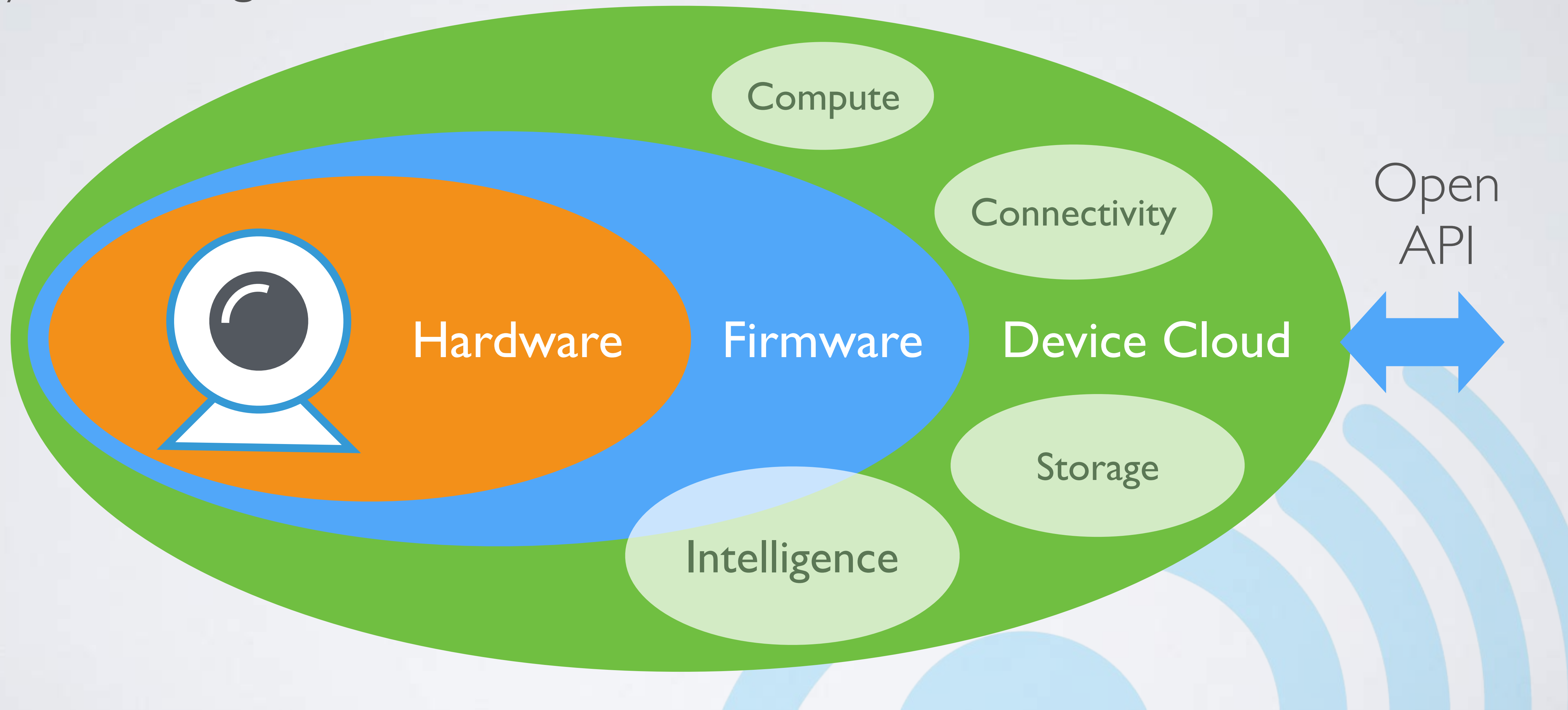
Device Cloud as a Component

Hardware vendors provide...



Device Cloud as a Component

System integrators need...



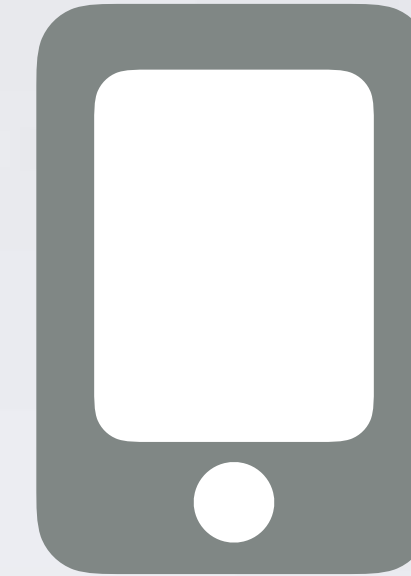
SKYWATCH / FET OPEN PLATFORM



Service Layer

Application & DB
Server Cluster

Billing
Access



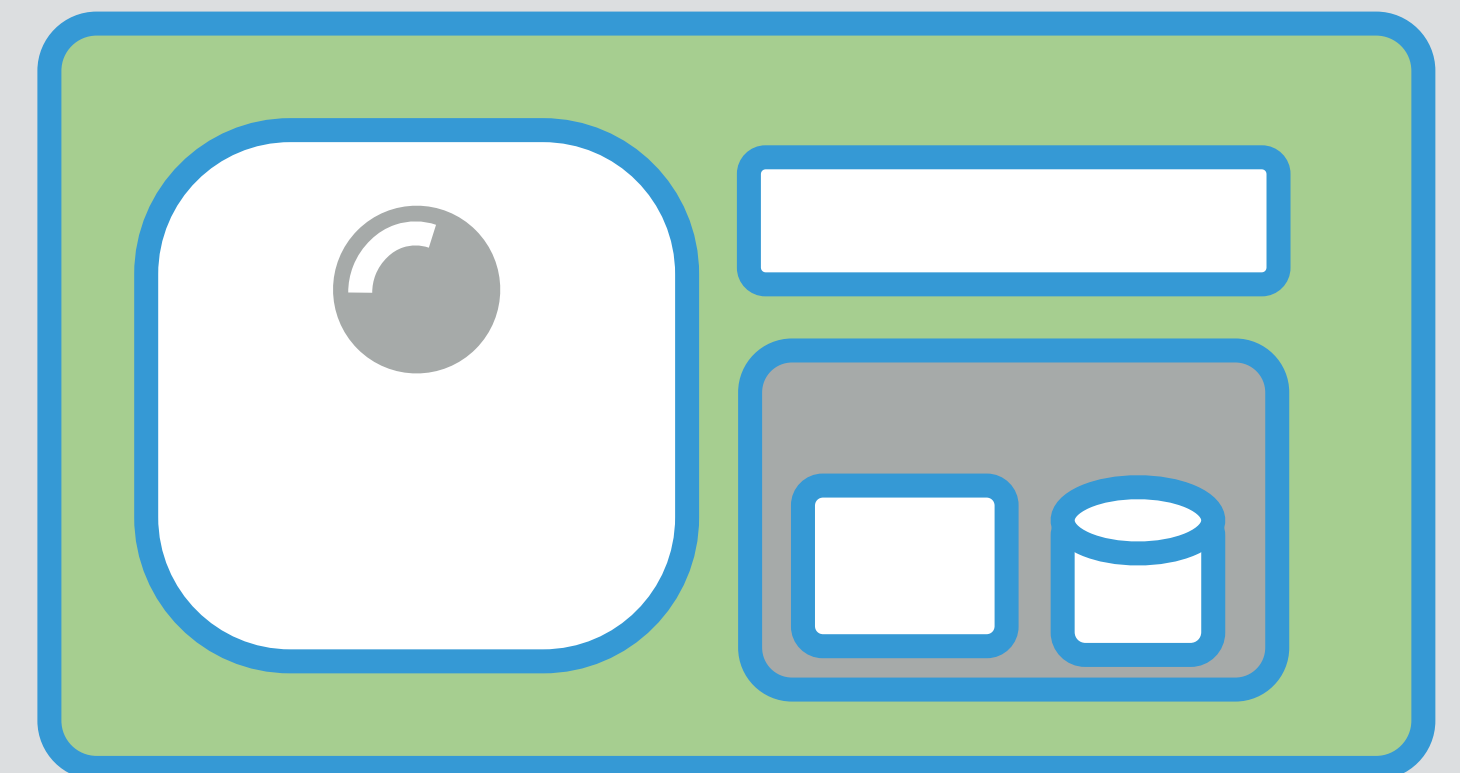
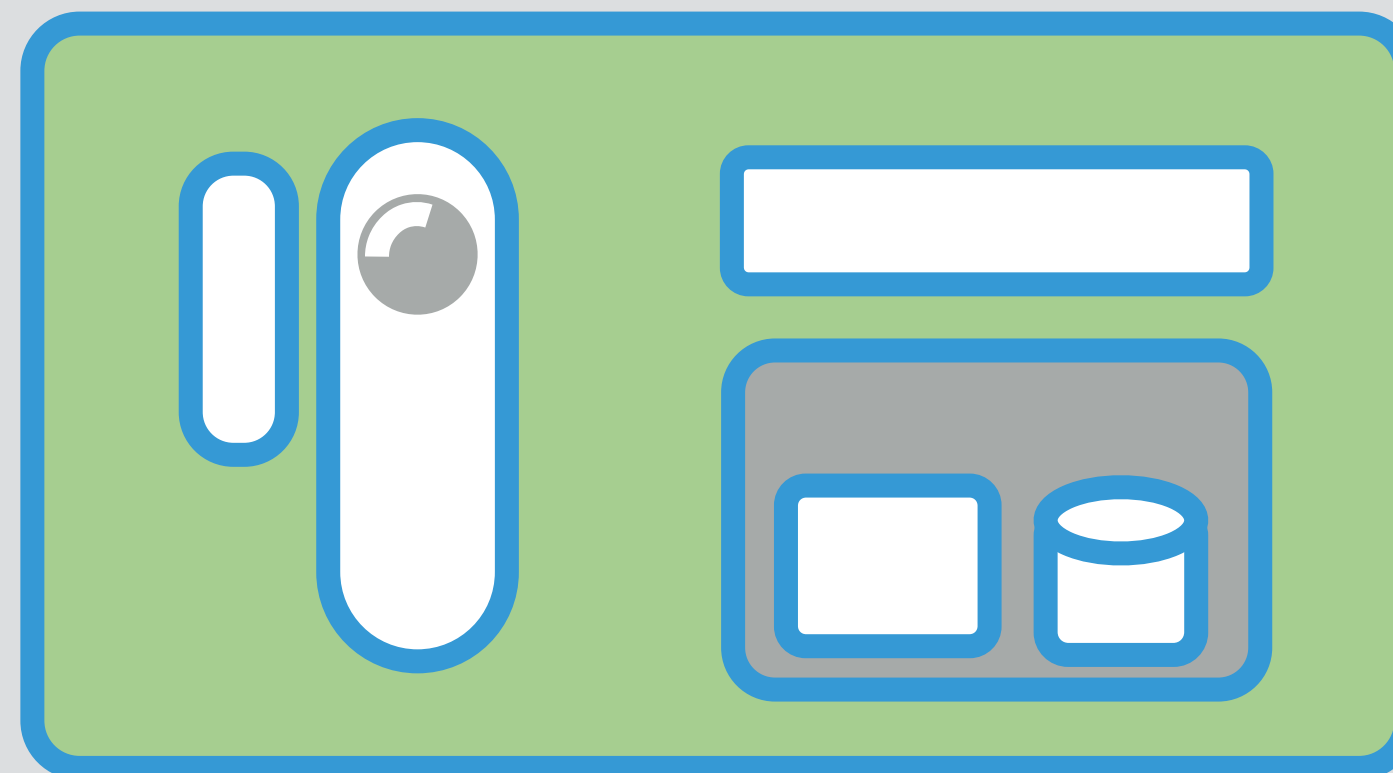
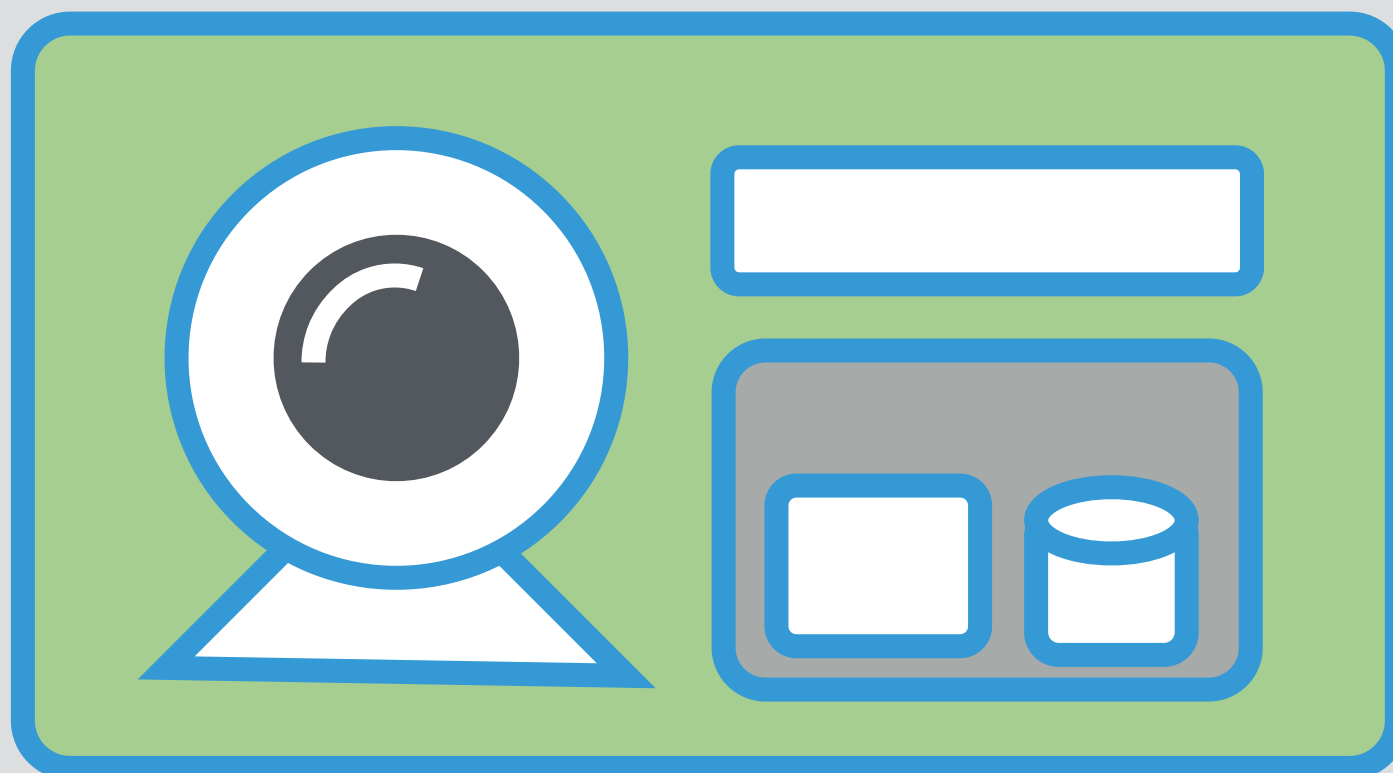
Processing Layer

Rule Engines

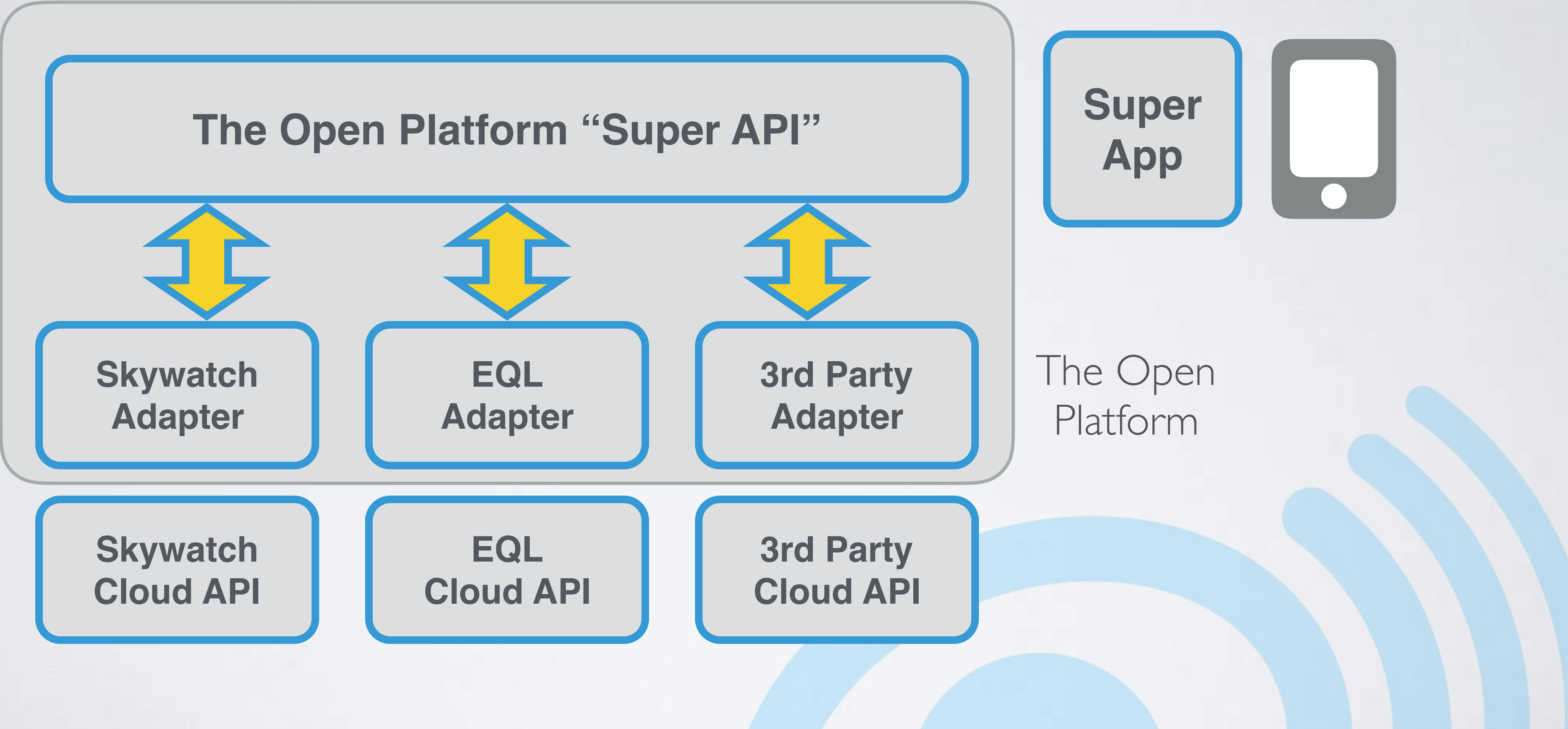
Analytic Engines

API Adapters

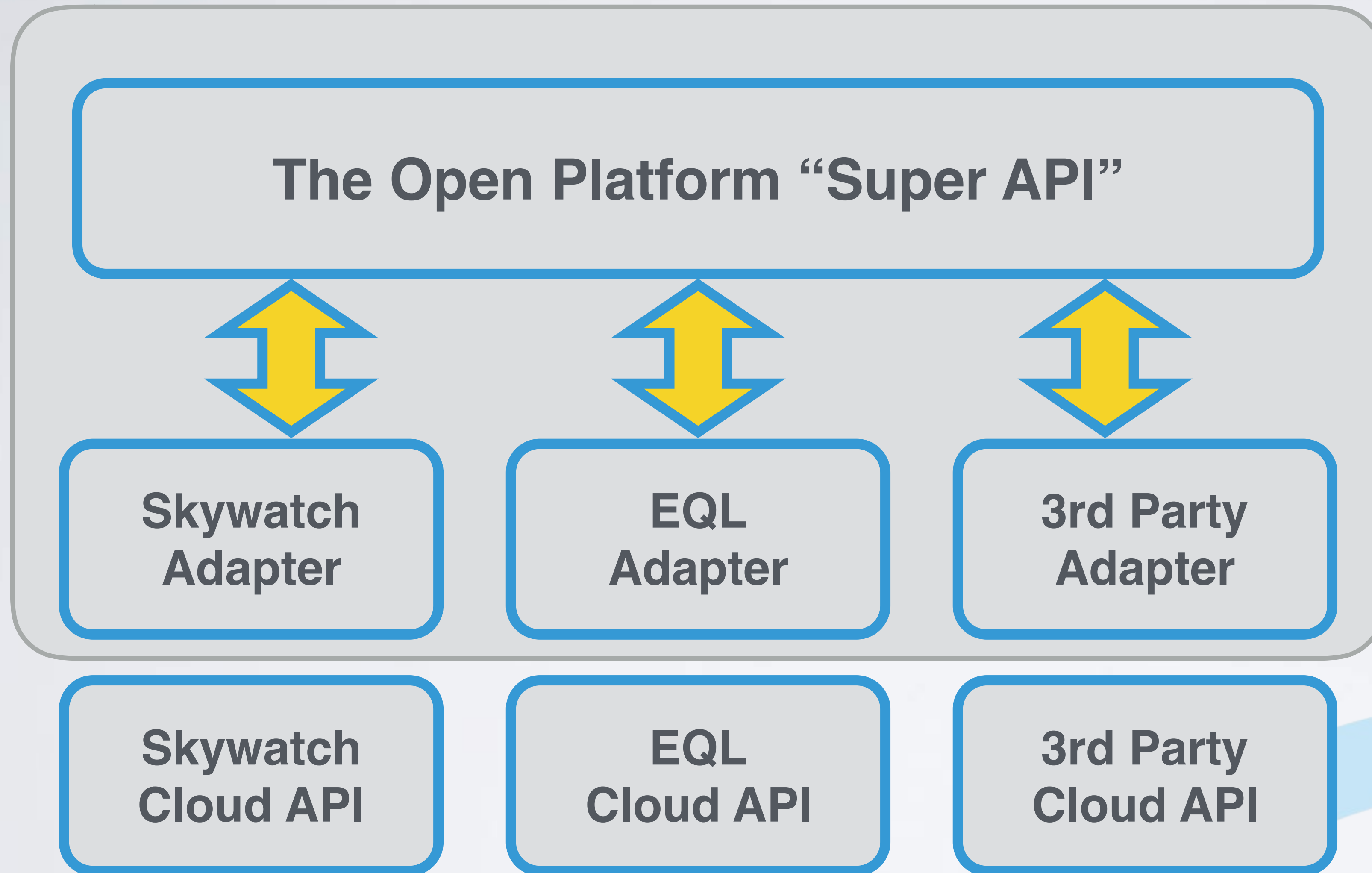
Connection Layer



Service Adapter for the Super API



Single Language/Skill for Smart Speaker



AWS
Lambda



Alexa smart
home skill



The Open
Platform

ALL WE NEED IS IMAGINATION
(THAT WE DON'T HAVE)





Shay Mitchell, actress

Waves or twists?
I say, why choose?

Pantene. Nature Fusion.

Forget trade-offs. Fusion is about getting all the style without all that damage, plus a touch of cassia too! The pro-vitamin formula works wonders and helps make hair stronger against splits and breaks.* So get creative. Your hair can take it.

* vs. non-conditioning shampoo.

PANTENE PRO-V

HEALTHY MAKES IT HAPPEN

Find your solutions at pantene.com



©2011 P&G



16 mg "tar," 1.1 mg nicotine av. per cigarette by FTC method.

SURGEON GENERAL'S WARNING: Cigarette
Smoke Contains Carbon Monoxide.



Ambience Butler

On-site communication system design for vacation rental guests

Ambience Butler enables convenient and friendly communication between vacation rental guests and their hosts. It is a virtual butler which exists amidst epaper information center, mobile messenger chatbot, and a collection of interactive pods. The epaper information center displays relevant information such as local weather, notes from the host, etc, with the help from distributed pods that can communicate with guests via light and sound. The pods are also designed to trigger based on time, presence or conversation between the guests and the host. The butler can also provide suggestion via messenger chatbot.

Host Guest Butler(Alfred)

Hi @Alfred, do you know how to use this oven?

Sure! Let me show you.



Welcome! Make yourself at home. If you have any question, @Bob is here at your service.

Hi I am Alfred, I am at your service, let me play some music from your home country first.

Would you like to visit the garden of the day? Come up here.

I have prepared some local desserts in the kitchen.



Discipline: Professional Concept



Skywatch Innovation

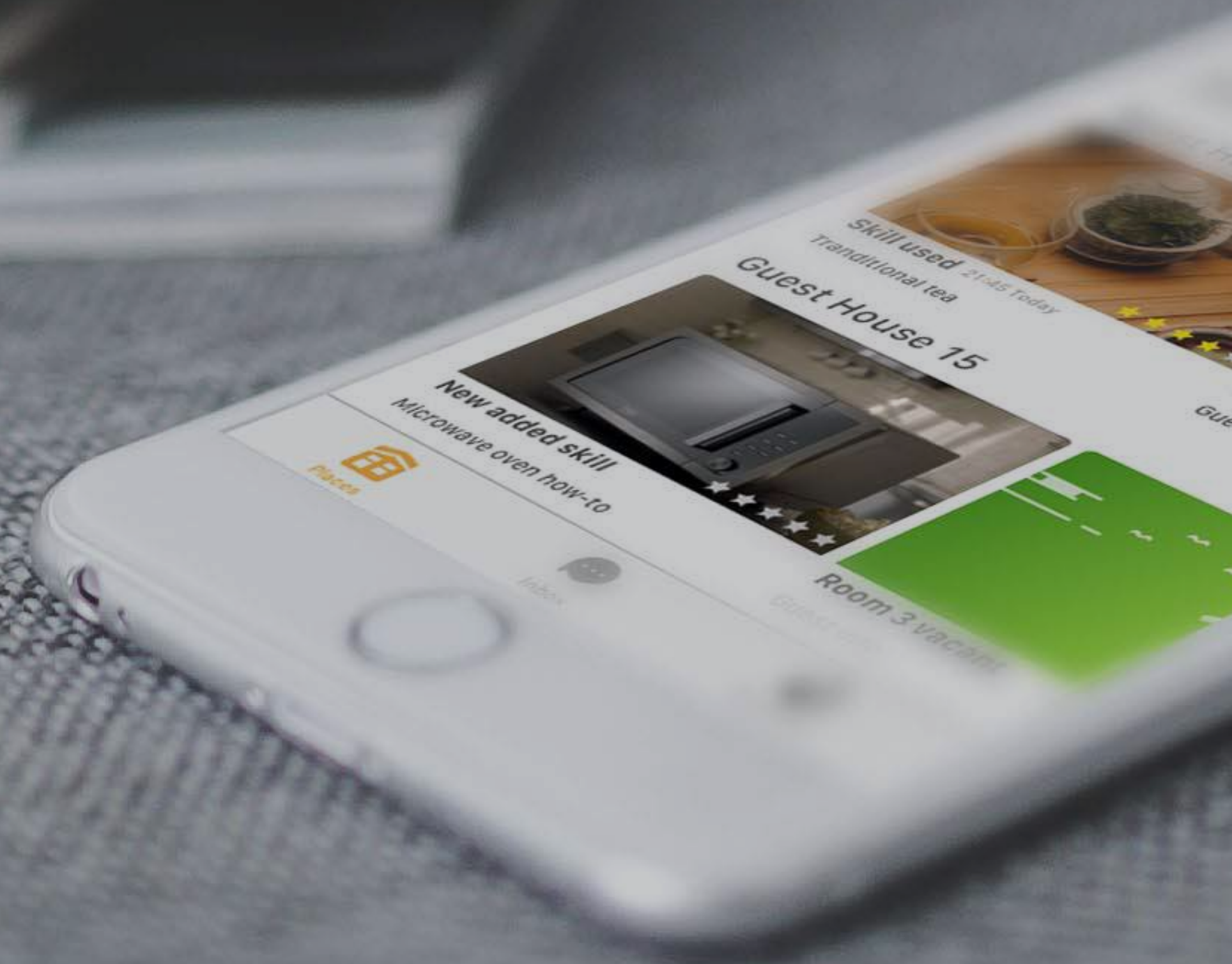
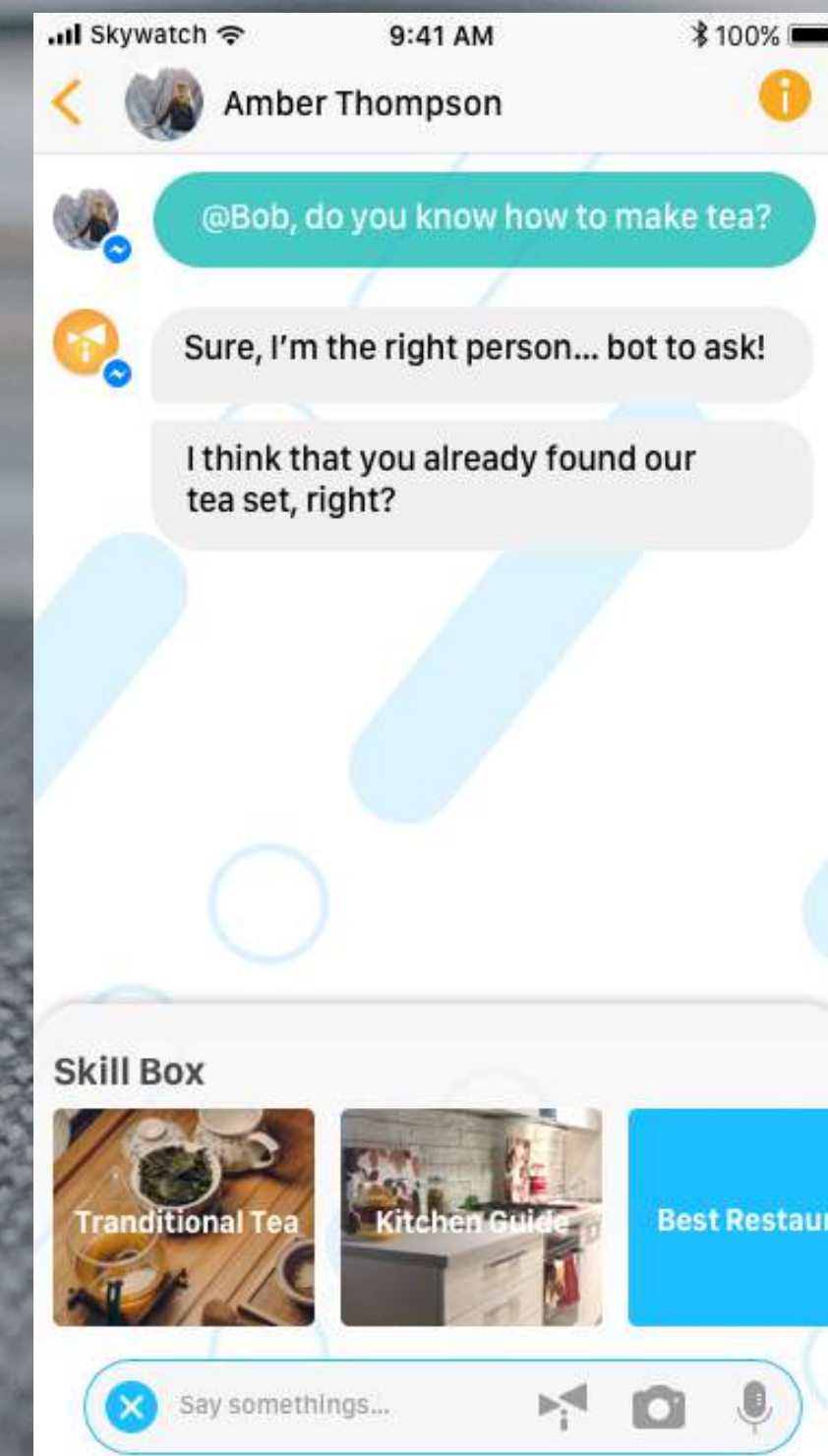
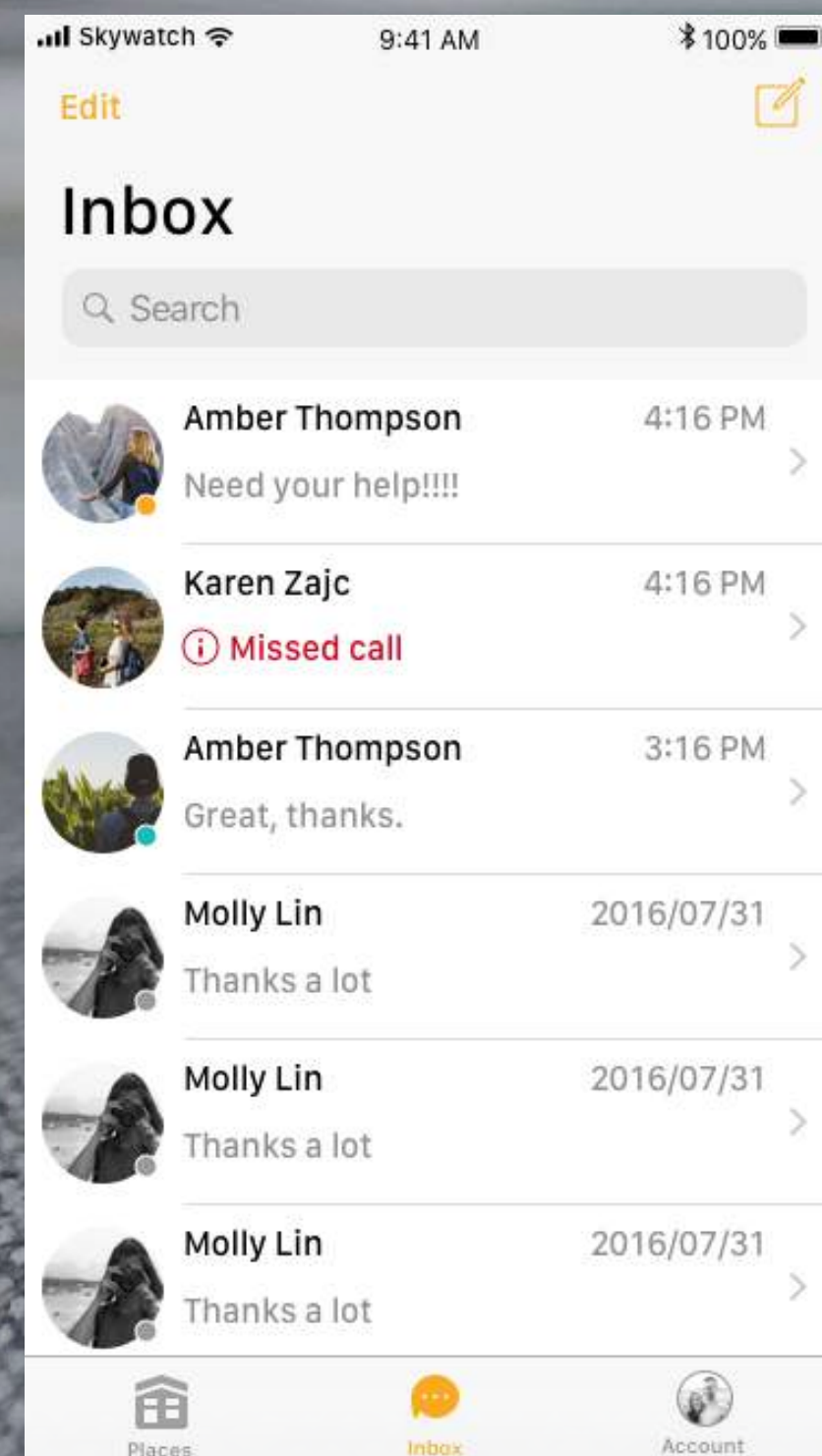
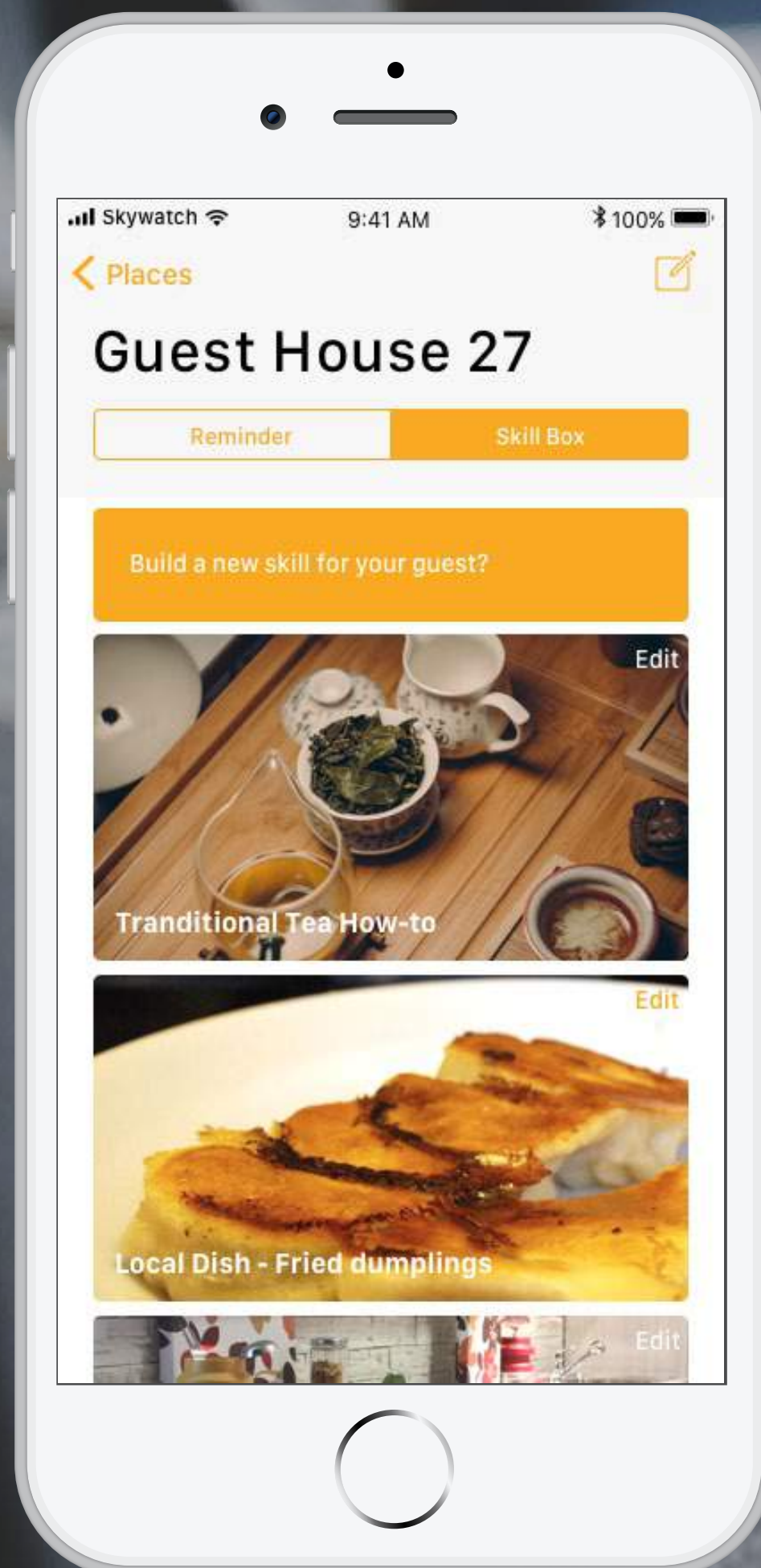
Skywatch
Taipei City, Taiwan

KuanYu Chen, PoHao Wang, WeiChao Chen

> GO TO PROFILE

Build Your Own Services

With **Ambiance Butler**, the host could serve the guests in a much better capacity. Simply build new skill sets for the Butler chatbot, such as house tutorials, local recommendations, etc., and let the AI technology take care of the customer services for the host.



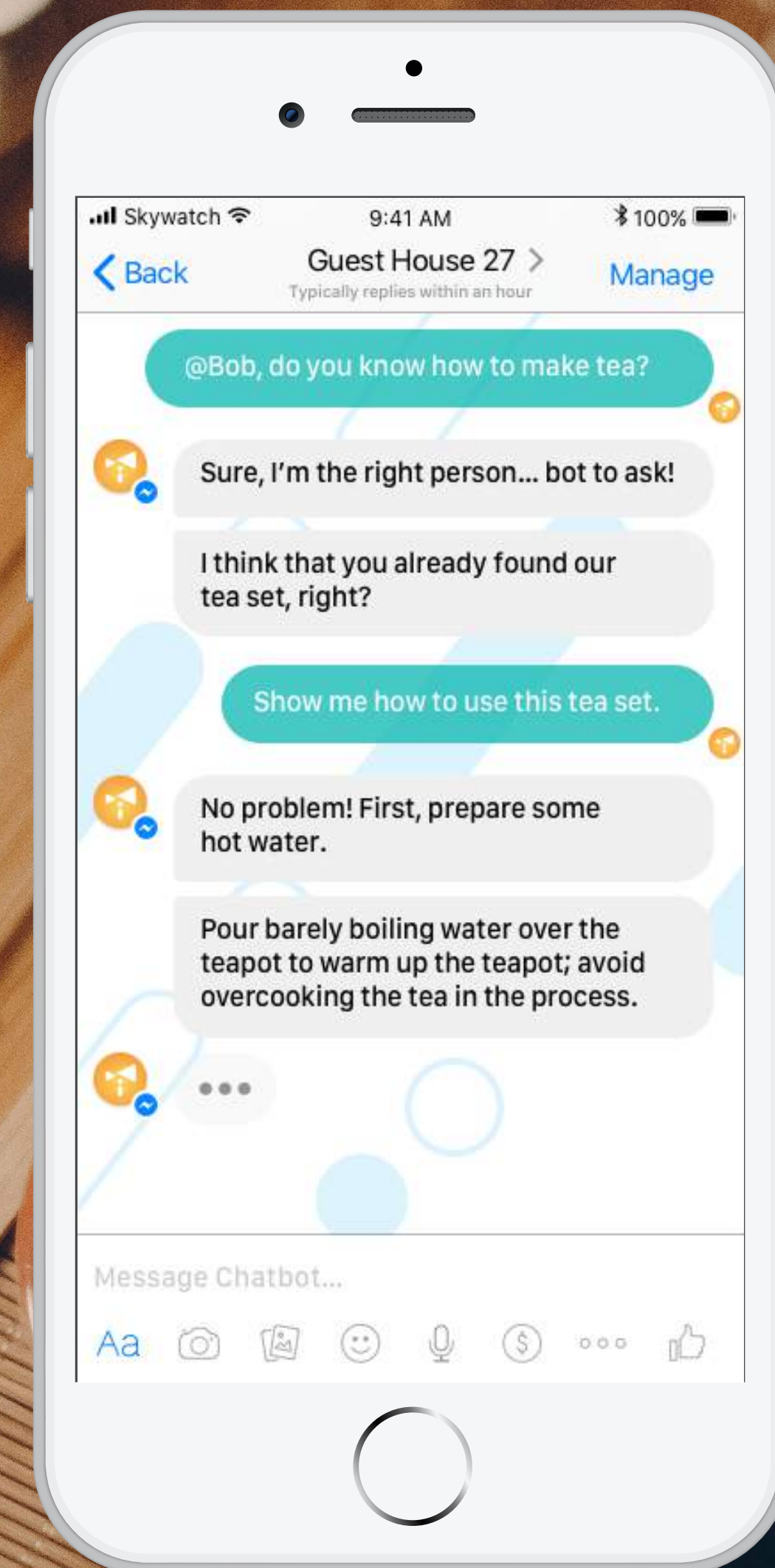
Personalized Welcome Messages

Greet your guest by playing welcome music from their home country. You can also set up personalized welcome messages, and the Ambience Butler can show them on the e-paper information display.



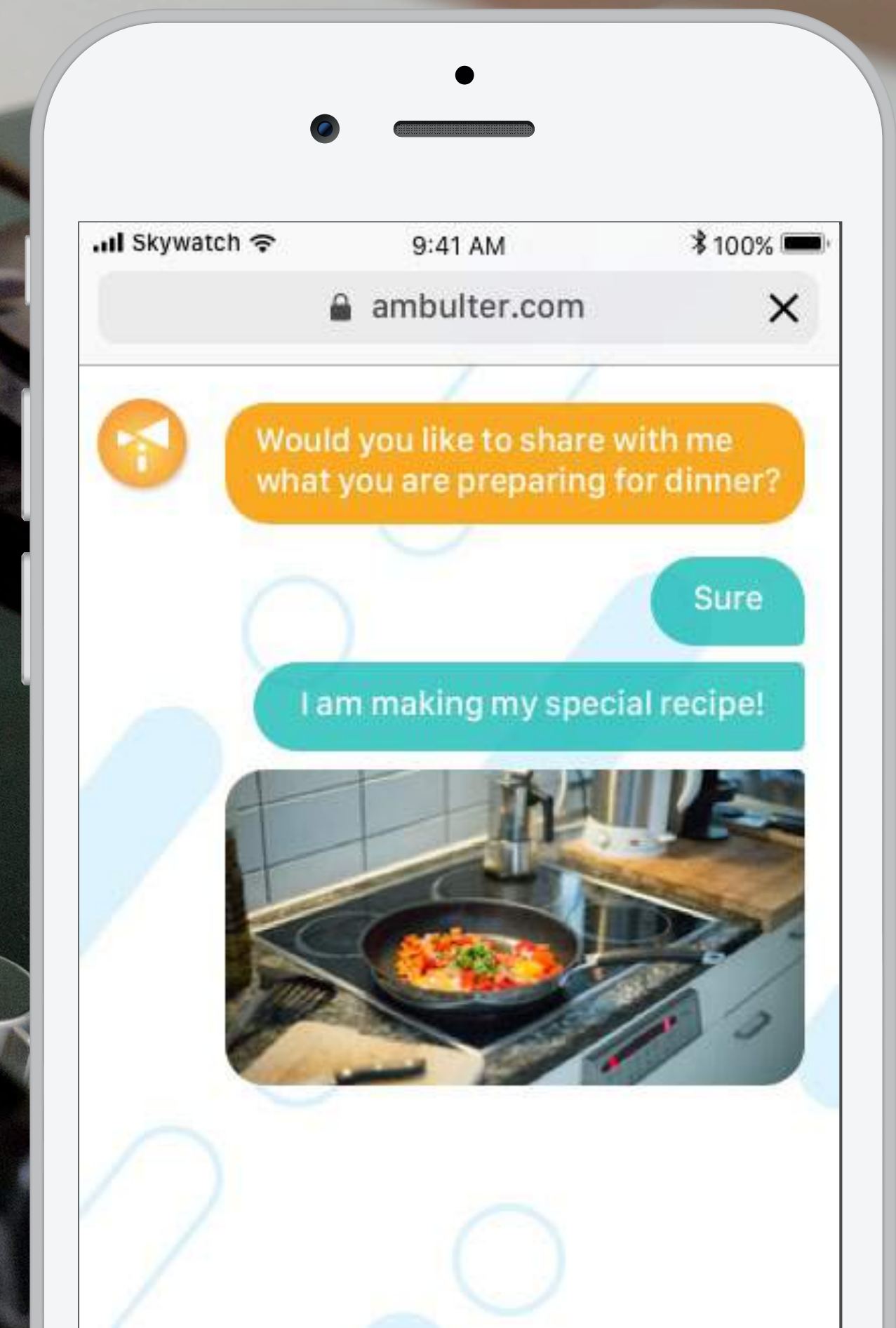
Interactive and Pervasive Guidance

During the stay, the guests can ask questions in the chat-room, such as how to operate a specific home appliance. Butler answers immediately, if it is within the range of the knowledge database, or it can notify the host through the mobile app for further assistance.



Preserve Memories of the Trip

Invite the guests to share events or fun activities that happened during their stay. Photos, recipes, or any comments can be preserved as special moments for the guests, and become sources of suggestions for future guests.



- The host introduces the Butler by inviting the guests into the chatroom.
- Butler provides digital keys, maps, and local information to the guests through the chatroom.
- The guests can make special requests from the chatroom.

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- The guests can make special requests from the chatroom.

Butler can help both the guests and the host, such as providing tutorials on home appliances to the guests, or notifying the host when there is too much noise.

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- Butler invites the guests to share experiences about their stay.
- Butler tells stories about the house, the host, and former guests.

- Butler invites the guests to share experiences about their stay.
- Butler tells stories about the house, the host, and former guests.

- The guests unlock the front door with a digital key.
- Butler greets the guests with music from their home country.
- Butler also notifies the host through the mobile app.

- The guests unlock the front door with a digital key.
- Butler greets the guests with music from their home country.
- Butler also notifies the host through the mobile app.

- Butler will make recommendations about the neighborhood, such as local restaurants.
- The guests can ask for recommendations as well.

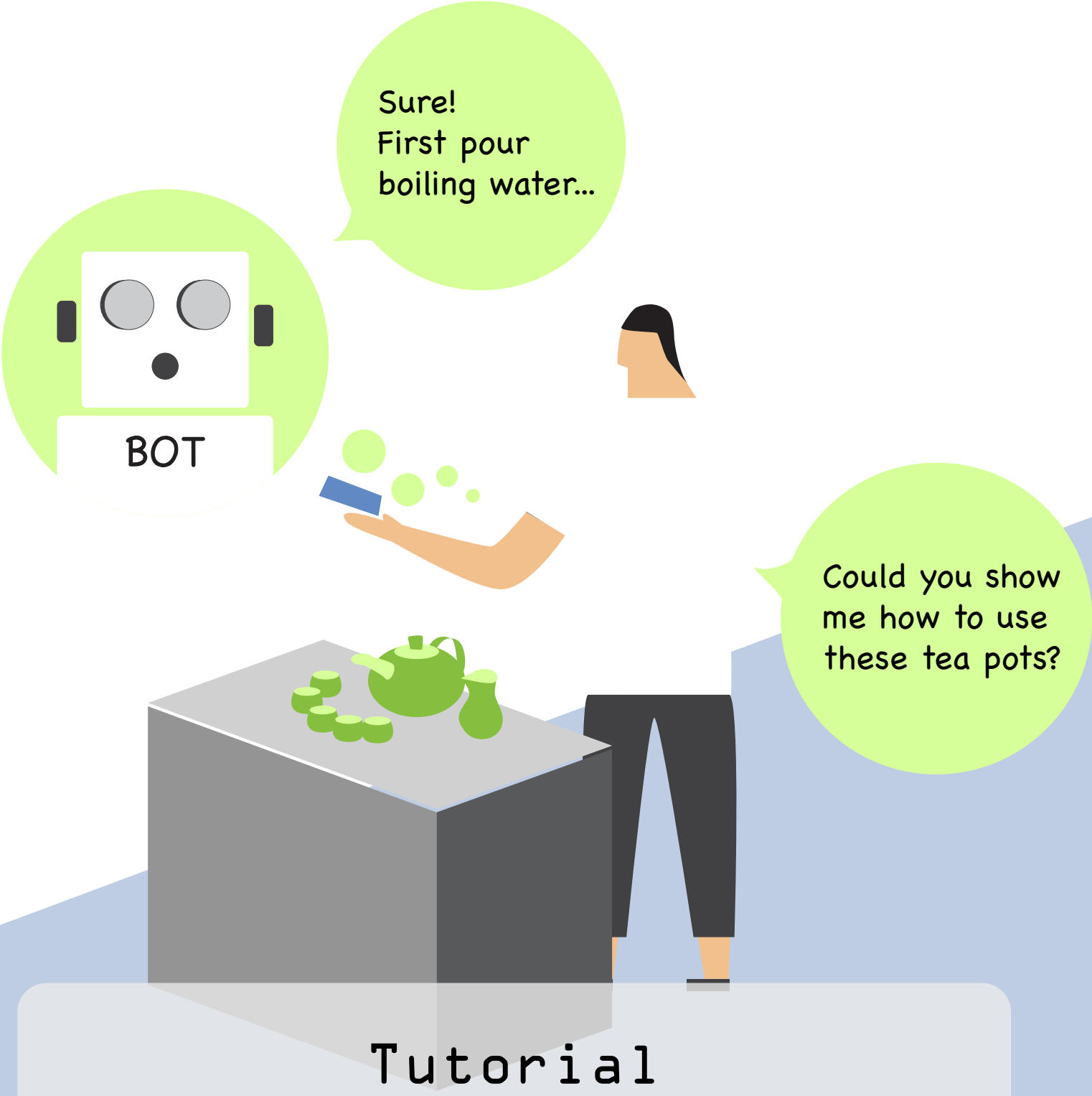
- Butler will make recommendations about the neighborhood, such as local restaurants.
- The guests can ask for recommendations as well.

After checking out, the Butler creates a movie from photos and videos shared by the guests and sends the movie as a souvenir to the guests

After checking out, the Butler creates a movie from photos and videos shared by the guests and sends the movie as a souvenir to the guests

Memory

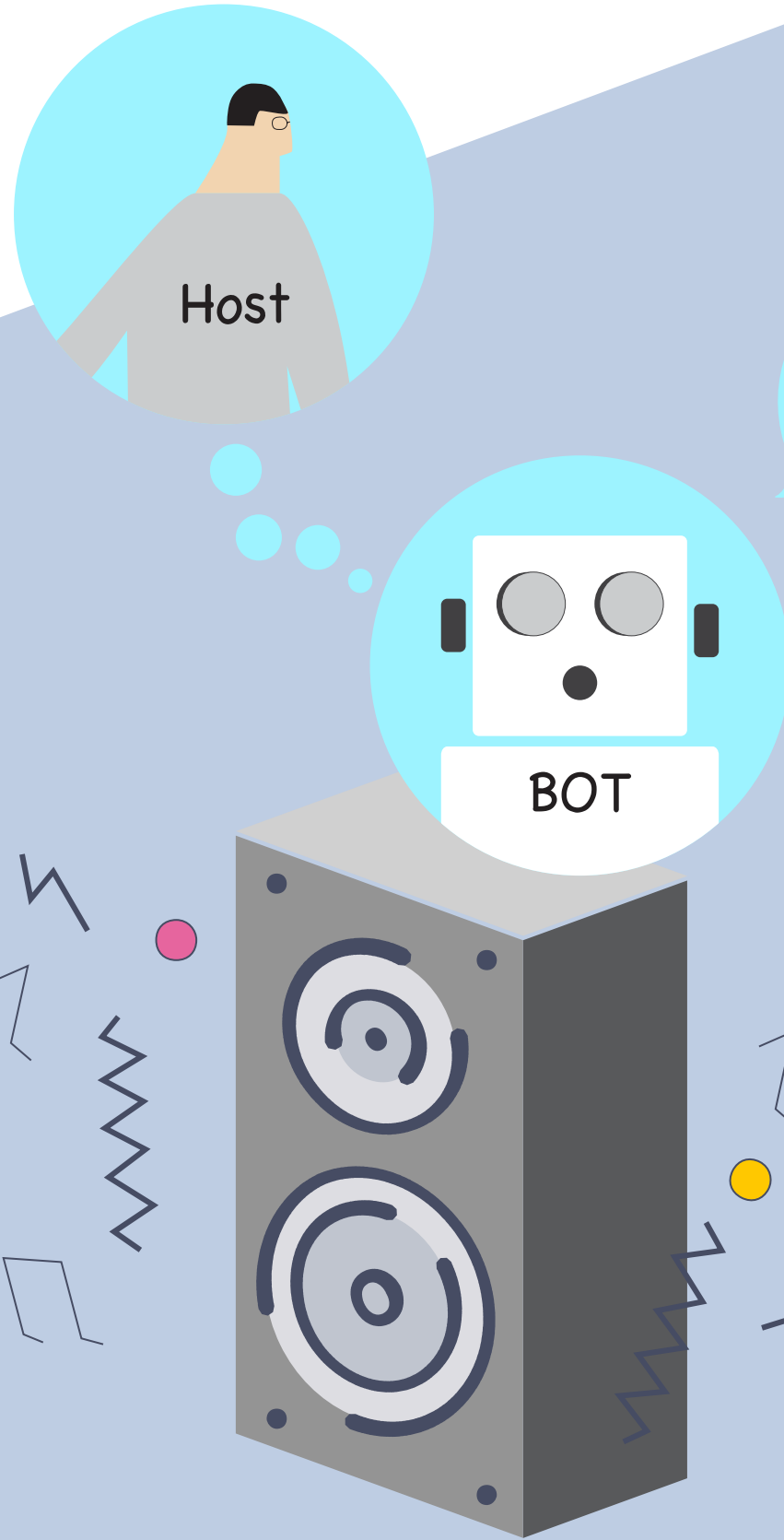
Provide Assistance



A diagram illustrating the 'Tutorial' function. On the left, a white robot head with two circular eyes and a small black dot for a mouth is labeled 'BOT' in a green circle. A green speech bubble from the BOT says, 'Sure! First pour boiling water...'. In the center, a person is shown from the waist up, holding a blue smartphone. To the right, another green speech bubble from the person says, 'Could you show me how to use these tea pots?'. Below the person is a grey rectangular table with a green teapot and several green teacups on it. The entire scene is set against a light blue background.

Tutorial

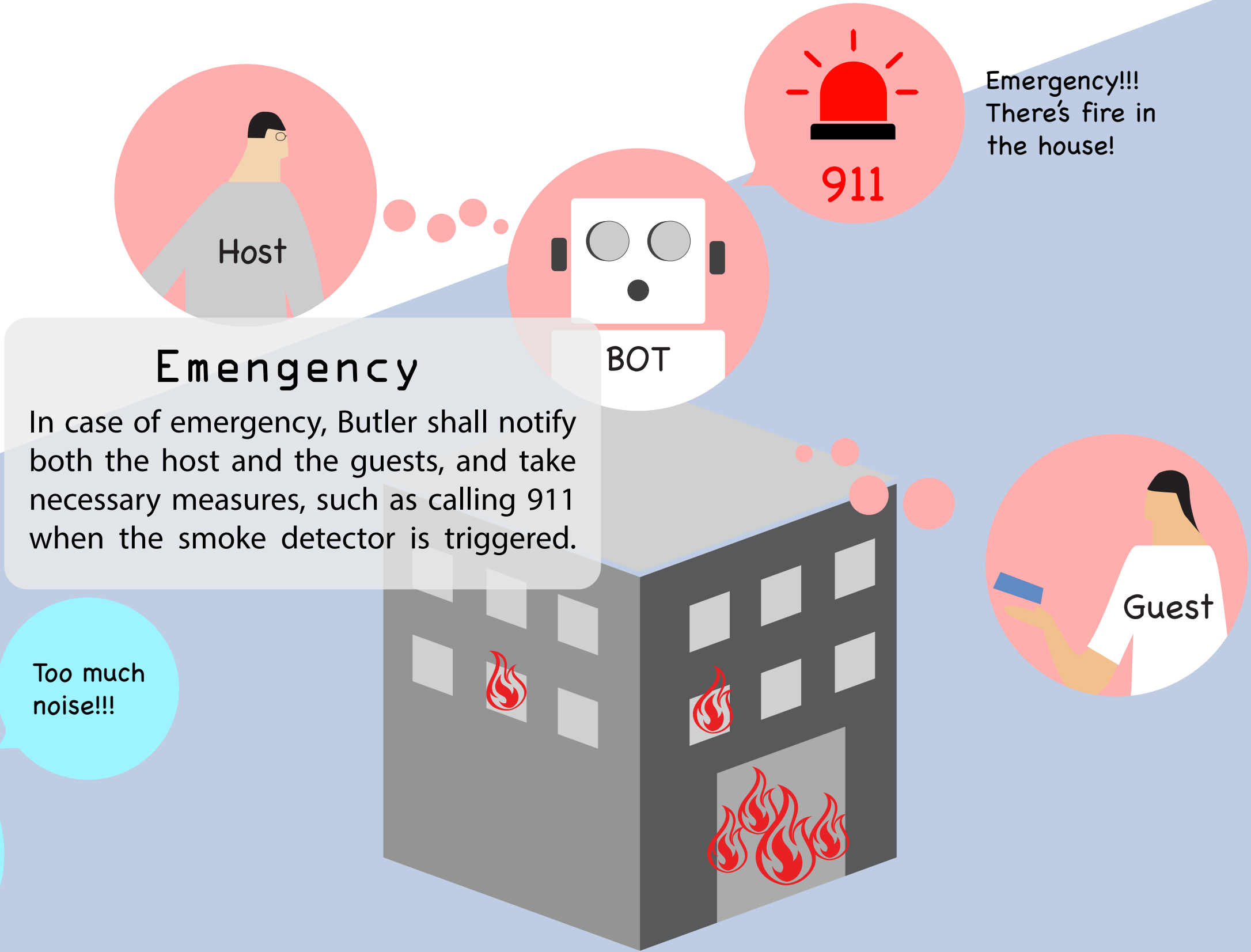
During the stay, the guests can ask questions in the chatroom, such as how to operate a specific home appliance. Butler answers immediately, if it is within the range of the knowledge database, or it can notify the host through the mobile app for further assistance.



A diagram illustrating the 'Rule Violation' function. On the left, a white robot head with two circular eyes and a small black dot for a mouth is labeled 'BOT' in a blue circle. A blue speech bubble from the BOT says, 'Too much noise!!!'. In the center, a person is shown from the waist up, holding a blue smartphone. To the right, another blue speech bubble from the person says, 'Could you show me how to use these tea pots?'. Below the person is a grey rectangular table with a green teapot and several green teacups on it. The entire scene is set against a light blue background.

Rule Violation

Butler notifies the host when the guests have violated the house rules. For example, if a decibel meter detects loud noise for a long period of time during late hours, Butler can notify the host.



A diagram illustrating the 'Emergency' function. On the left, a white robot head with two circular eyes and a small black dot for a mouth is labeled 'BOT' in a red circle. A red speech bubble from the BOT says, 'Emergency!!! There's fire in the house!'. In the center, a person is shown from the waist up, holding a blue smartphone. To the right, another red speech bubble from the person says, 'Emergency!!! There's fire in the house!'. Below the person is a grey rectangular table with a green teapot and several green teacups on it. The entire scene is set against a light blue background.

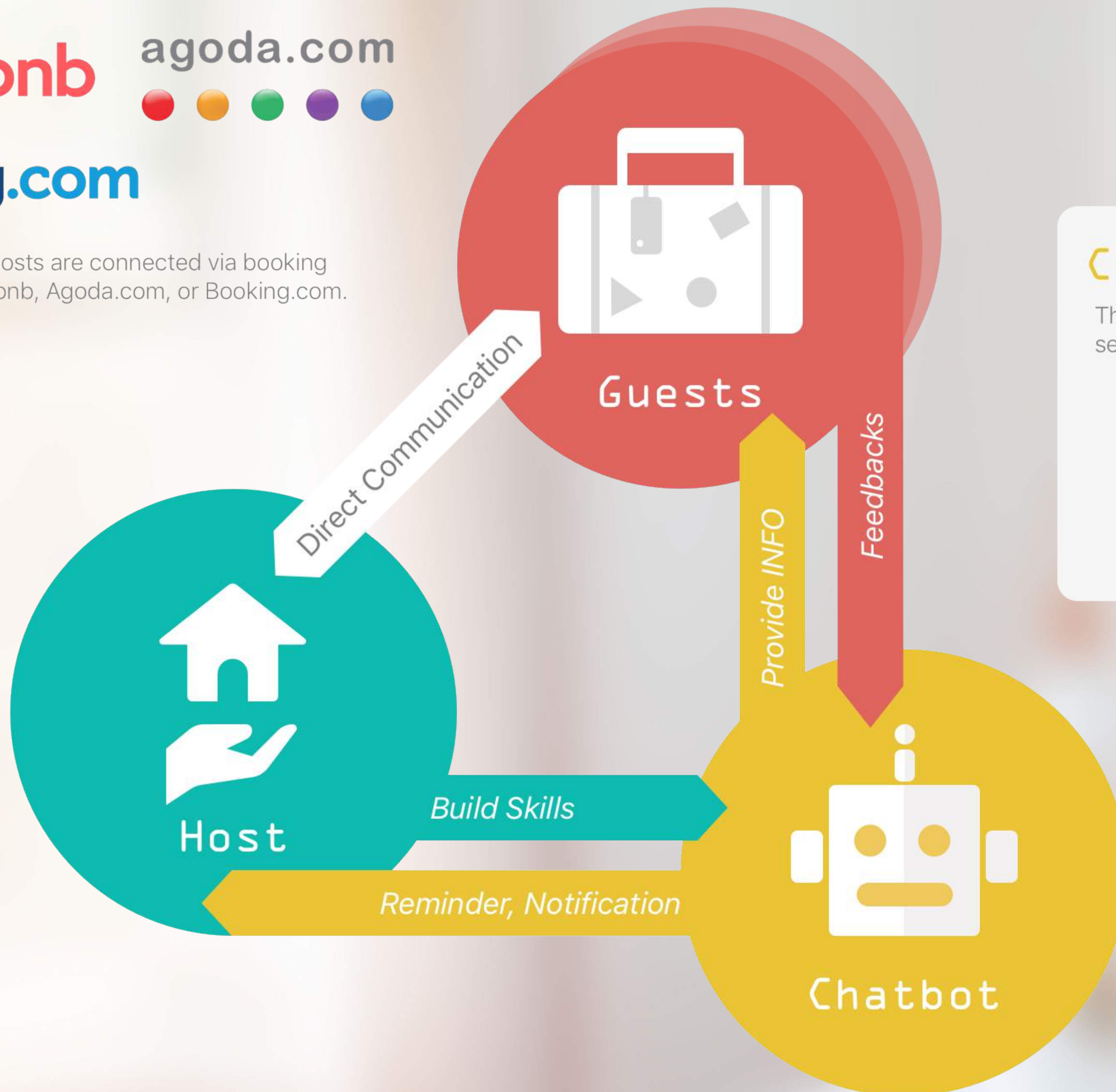
Emergency

In case of emergency, Butler shall notify both the host and the guests, and take necessary measures, such as calling 911 when the smoke detector is triggered.



Booking.com

The guests and the hosts are connected via booking services, such as Airbnb, Agoda.com, or Booking.com.



Connect to local services

The guests can access local services such as taxi, delivery services, and booking systems, via chatbot.



WhatsApp



facebook

The chatbot supports major messaging and social network platforms.



Smarter
Edge Devices

- + Edge computation
- + Lower bandwidth
- + Higher device value



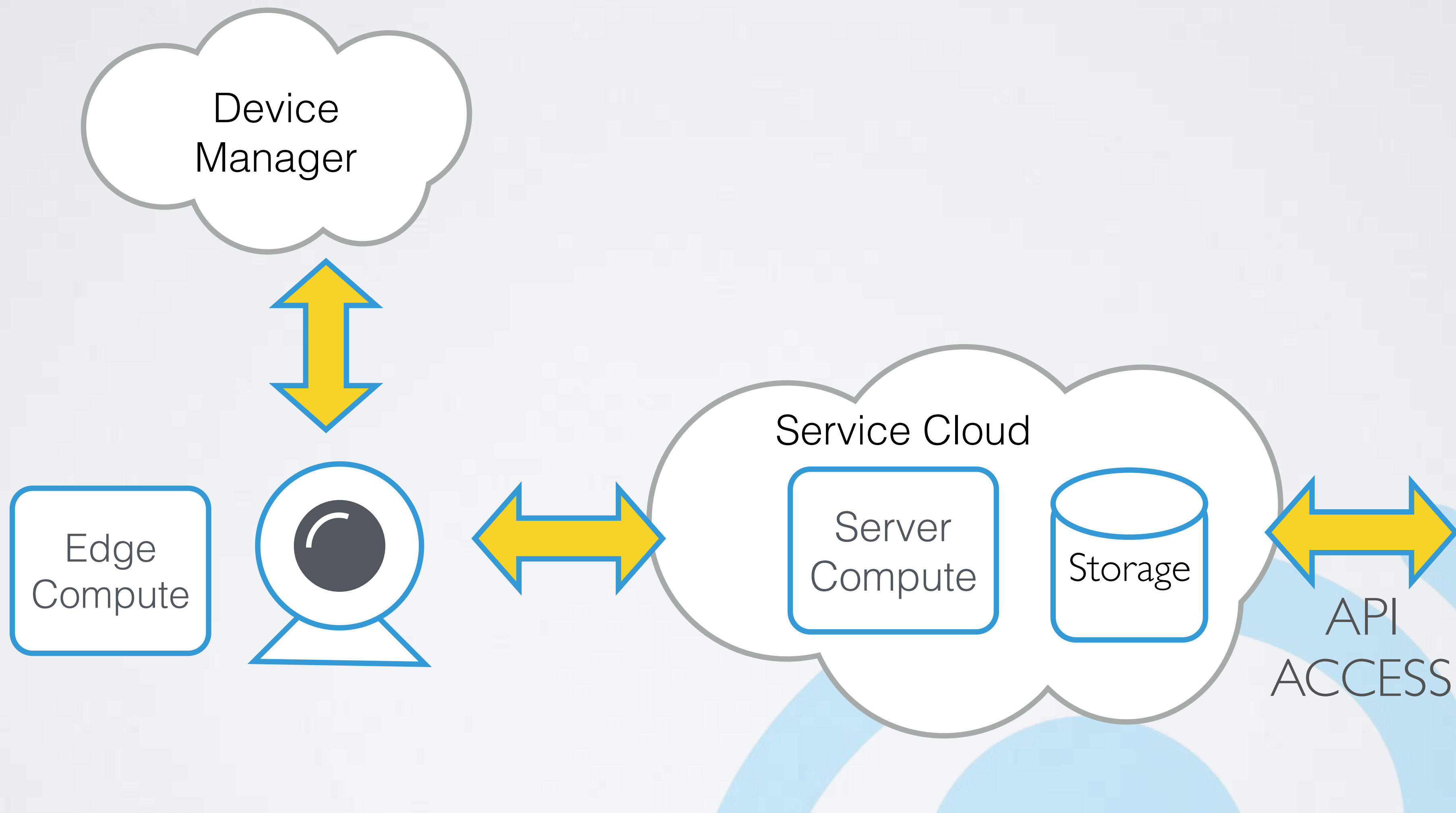
Cloud-Edge
Collaboration

- + Smaller cloud
- + Better privacy
- + Data democracy



Distributed IoT Intelligene

CLOUD MANAGED SMARTNESS



CLOUD MANAGED SMARTNESS

“Boosting stage”

ARM/Android SOC
General purpose
Inference engine
Configurable codec

Edge
Compute



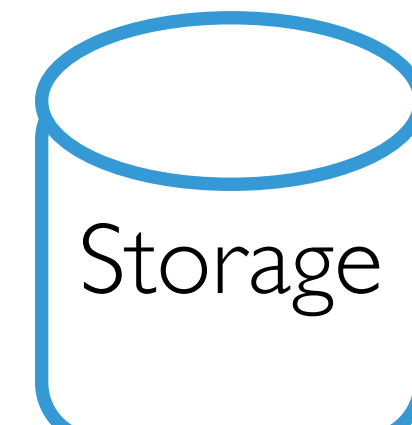
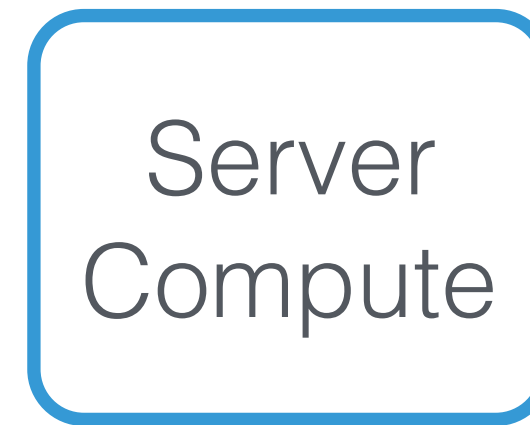
“Learning stage”

GPU friendly
Online learning
Encrypted domain analysis
Data collection / exchange

Service Cloud

Server
Compute

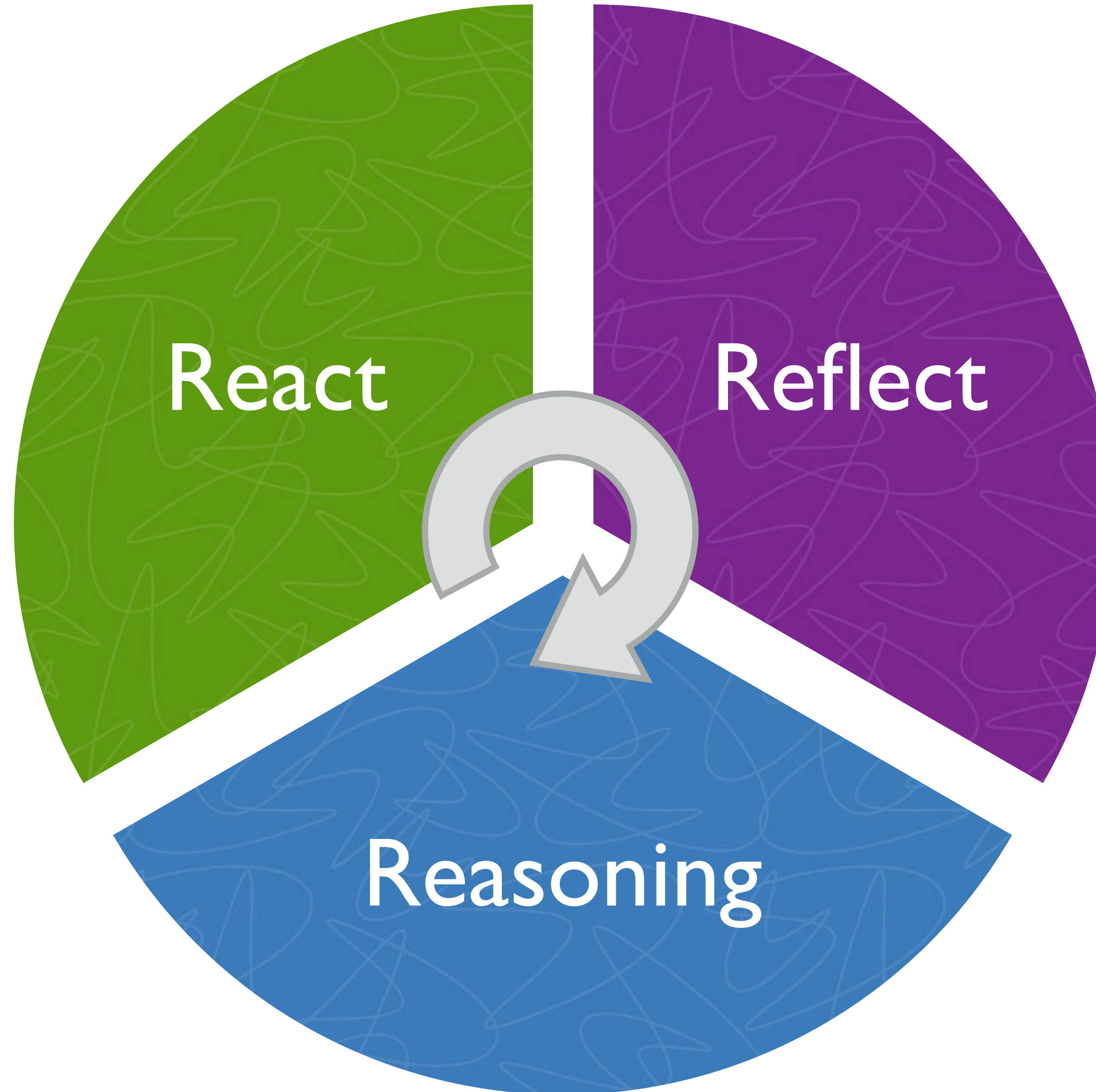
Storage



Phases of IoT Services

Immediate

Alarm
Notification
Single-device



React

Reflect

Reasoning

Short-term

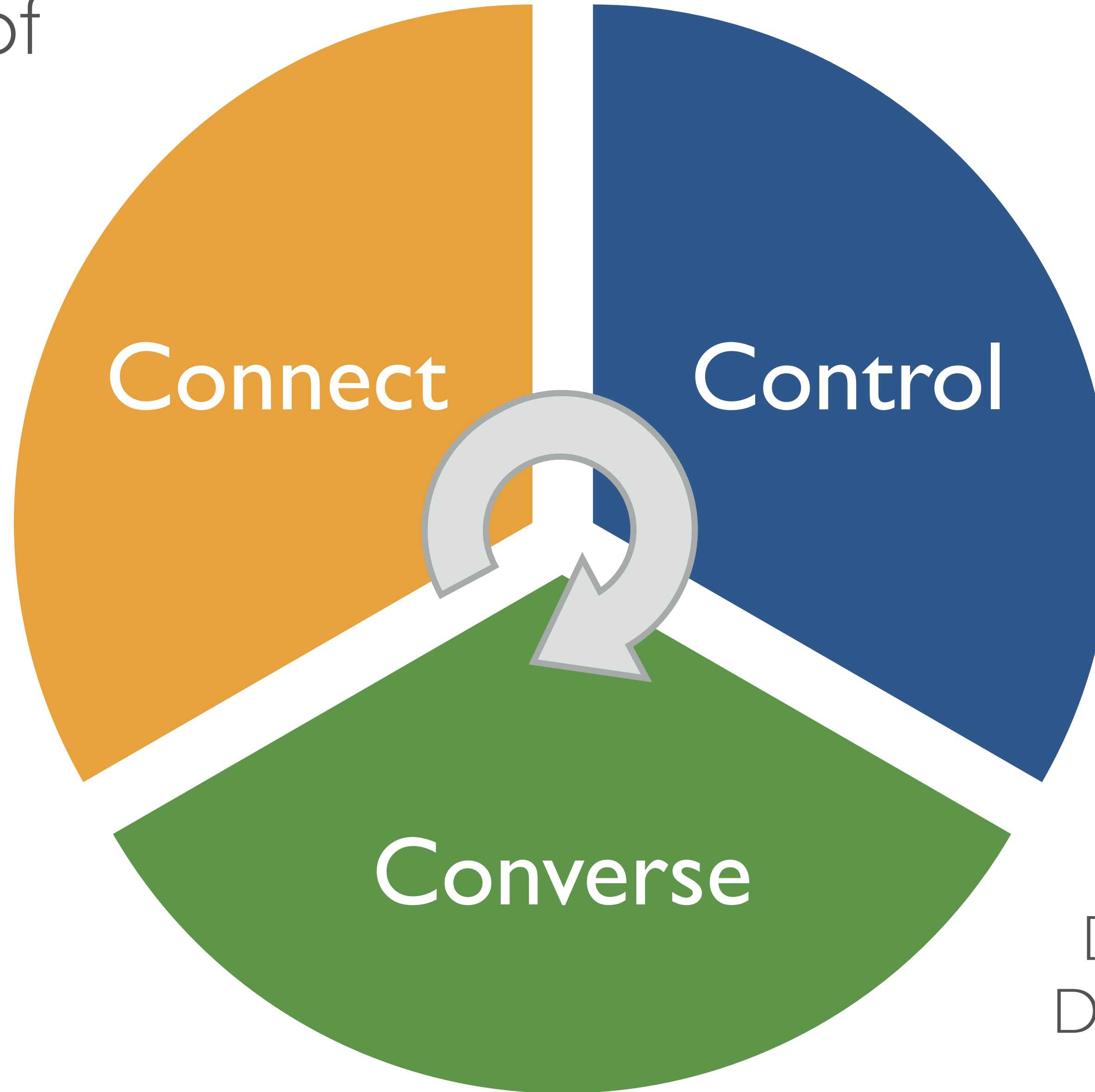
Smart reaction
Deduction
Cross-device

Long-term

Data analytics
Statistics
Cross-device and user

Sophistication of
IoT devices

Physical
Power consumption
Range & reach
Compatibility



Application
Remote access
Data upload
Security

Intelligence
Device to devices
Device to databases
AI and reasoning

Eiffel Tower
Robert Delaunay



Abstraction or Instantiation on Building the Internet for Smarter Places

A Keynote for PCaPAC 2018
by Wei-Chao Chen @ Skywatch
weichao.chen@skywatch24.com